“Form and Function Dental Clinic Integrated System”

A System Study Presented to

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In Partial Fulfillment

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1 INTRODUCTION

1.1 Purpose

The purpose of this user requirement document is to capture and document the needs and expectations of the end-users and stakeholders for the development and integration of the Patient Record, Appointment, and Prescription Management System for Form and Function Dental Clinic. This document will serve as a foundation for subsequent design and development activities, ensuring that the final product meets the users' needs.

1.2 Project Background

Form and Function Dental Clinic faces significant challenges in running its business efficiently and providing optimal patient care. One major issue is the need for a dedicated system for the clinic, as they rely on different applications for various tasks. This fragmented approach creates inefficiencies and hampers coordination. The clinic must also work on effective communication, complex billing and invoicing procedures, appointment scheduling issues, and manual record-keeping. Managing records is time-consuming and prone to errors, mainly since the clinic still relies on paper-based systems. Manual appointment scheduling leads to clarity, double bookings, and efficient time use for dental professionals. Furthermore, the outdated paper-based data recording method poses risks of errors and lost records.

1.3 Business Objectives

The business objectives for the Form and Function dental clinic project are as follows:

- To create a system that centralizes all clinic operations, improving coordination and effectiveness for a seamless patient experience.

- To transition from a paper-based to a digital system for accurate, secure, and easily accessible patient records.
- To reduce double bookings, minimize wait times, and maximize dental professionals' time.

1.4 Project Scope

Our innovative dental clinic system is designed to transform how dental practices operate and communicate with patients. With a user-friendly interface and a range of features, our system aims to optimize clinic operation and elevate patient care.

The primary functions to be provided by the system are listed below:

- User login and registration forms for doctors and assistants
- Ability to edit or scan patient information.
- Patient Record Management
- The "Contact Us" section lets patients contact doctors or assistants quickly.
- Option to save patient records as PDF, Excel, and CSV for easy printing
- Book Appointment Functionality
- Patient Data Autofill Button for Prescription

1.5 Project Objectives

The project is to:

- To create a unified system to address the inefficiencies of using multiple apps.
- To create patient records for a comprehensive overview to print.
- To create an appointment form for patients.
- To facilitate appointment creation from the doctor's or assistant perspective in case they haven't scheduled online.

- To create slot availability to allow easy scheduling.
- To implement a payment feature where users can input bill amounts, make payments, and receive change.

2 Identified Risks, Assumptions, and Constraints

The following are the identified risks, assumptions, and constraints related to requirements collected at this stage from the business users and/or the stakeholders involved.

2.1 Identified Risks

RISK	DESCRIPTION	MITIGATION
Data Security Risks	Patient data is sensitive and must be protected from unauthorized access or data breaches.	● Implement strong access controls and user authentication mechanisms. ● Establish protocols for handling sensitive information and educate employees on data security procedures.

Regulatory compliance	Dental clinics should consider the Data Privacy Act of 2012 (Republic Act No. 10173) and its implementing rules and regulations, which govern important privacy and security laws. Legal and financial consequences may occur from a failure to abide by privacy and Security requirements.	● Stay informed about relevant regulations and ensure the system is designed and implemented in compliance with them
System Reliability and Performance	The dental clinic system may experience reliability or performance issues, leading to downtime, slow response times, or system unavailability during peak usage.	● Ensure the system is built on reliable hardware and software infrastructure to minimize the risk of failures and maximize performance. ● Establish backup and recovery mechanisms to minimize downtime and restore system functionality during failures.

2.2 Assumptions

ASSUMPTION	DESCRIPTION	MITIGATION
Compatibility with Different Devices and Operating Systems	The dental clinic system assumes compatibility with various devices (computers, tablets, smartphones) and operating systems (Windows, macOS, iOS, Android) commonly used by dental clinic staff and patients.	● Conduct extensive testing on different devices and operating system combinations to identify and resolve compatibility issues.
Optimal System Performance and Scalability	The dental clinic system assumes it can deliver optimal performance and scalability to handle the expected user load, data volume, and system usage without degradation or limitations.	● In case of an emergency or system breakdown, establish a disaster recovery plan setting out the procedures for restoring systems.

2.3 Constraints

CONSTRAINTS	DESCRIPTION	MITIGATION
User Adoption and Training	The effective functioning and utilization of the dental clinic system rely on staff and patients' acceptance and proficient usage of the system. Nevertheless, any reluctance from users or insufficient training could hinder the system's acceptance and utilization.	● Provide user-friendly interfaces with intuitive navigation and instructions to facilitate ease of use and minimize the learning curve. ● Inform and educate dental clinic staff and patients on how to use the system, its features, and its benefits.
Data Migration and Conversion	The dental clinic system may encounter constraints when migrating and converting data from existing systems or formats to the new system, which can impact the data's accuracy, completeness, and integrity.	● Conduct a thorough data assessment and analysis to understand the existing data's structure, quality, and format. ● Cleanse and normalize the data, ensuring consistency and accuracy, before initiating the migration process.

3 Proposed System Overview

3.1 Description of the Proposed System

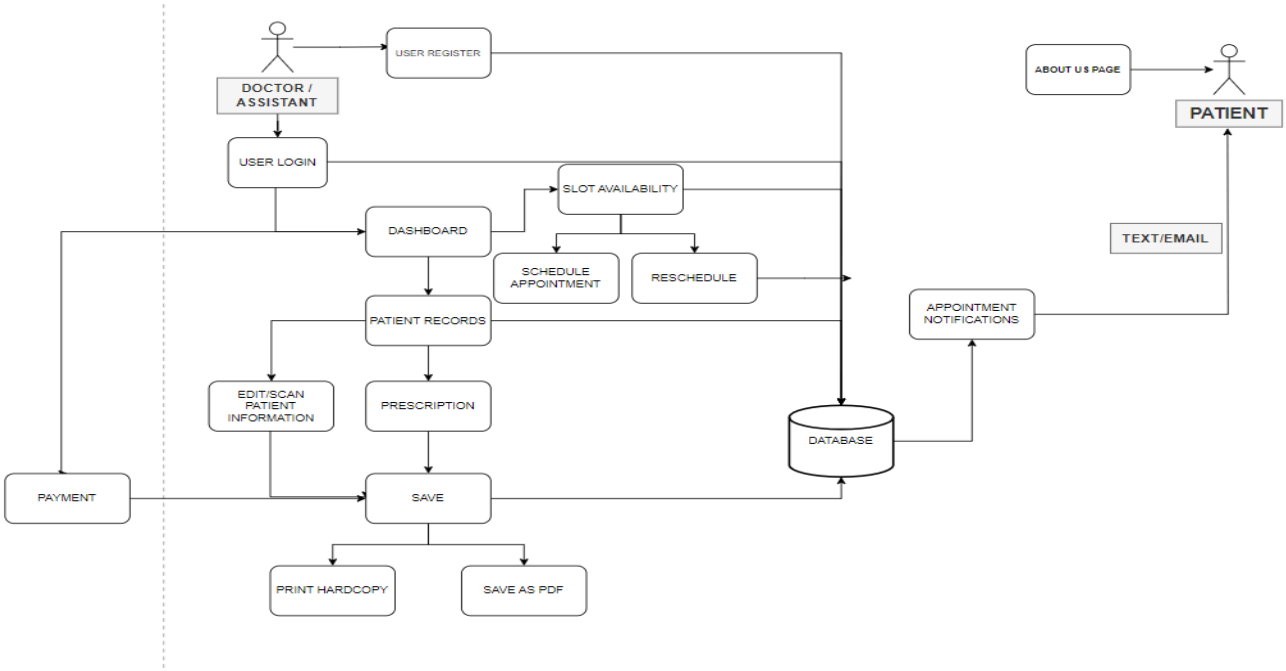


Figure 1.

In Figure 1, the process starts with registration and login; the doctor will have access to the dashboard upon logging in. The doctor can perform actions such as scheduling or rescheduling an appointment. The patient records can be updated and accessed, including edited information. There is also a payment process: the billing or payment for the medical services. The patient can receive appointment notifications via email or text.

3.2 System User Profile

User Role	Responsibilities
System Administrator (DOCTOR)	Responsible for operating the whole system.

System Administrators (ASSISTANTS)	Responsible for operating the whole system if the doctor is not around.

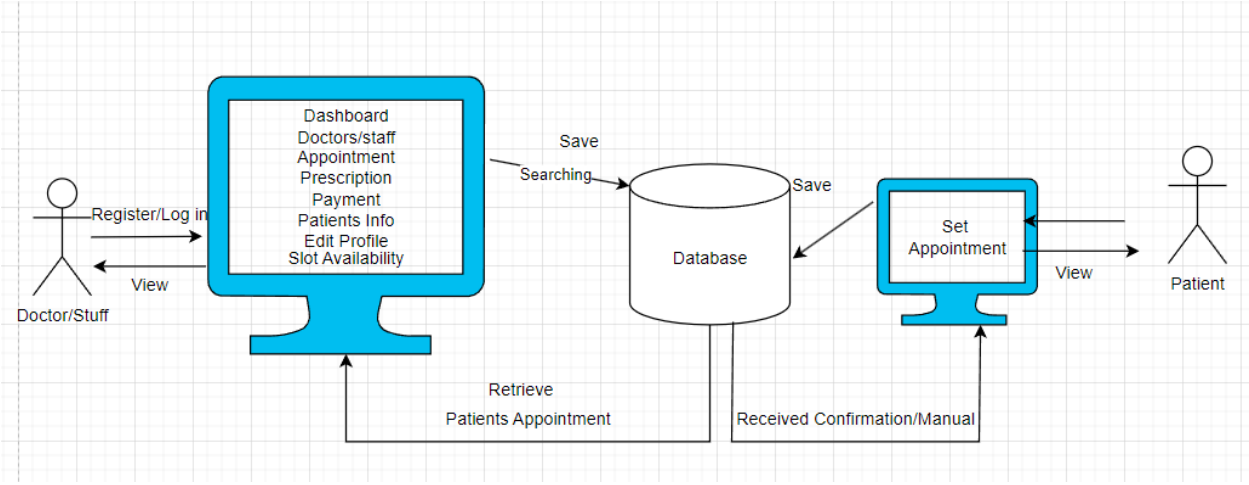


FIGURE 2.

3.3 Data Flow Diagram

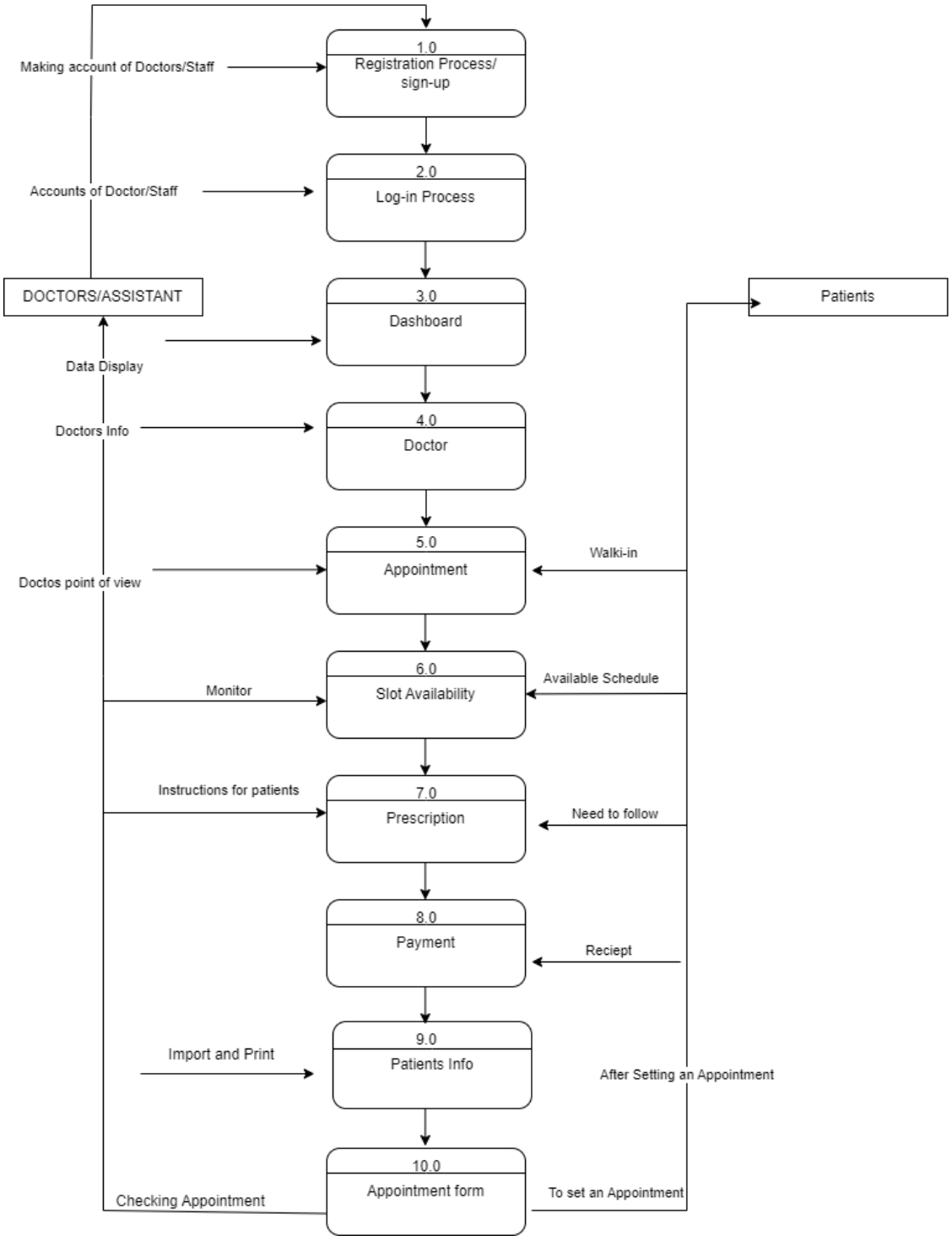


FIGURE 3.

In this diagram, the first step is the registration process, the second step is the login process, the third step is updating the account, the fourth step is scheduling an appointment, the 5th step is payment, and the last step is printing the invoice. It also indicates that a rescheduling is a possibility.

4 Architectural Design

4.1. High-level Components and Interfaces

4.1.1 Components

Doctor Component: The doctor component facilitates administrative functions and tasks tailored to doctors. It is intended for users with doctor-specific privileges, enabling them to oversee and manage pertinent aspects of the system.

Patient Component: This component enables patients to book appointments seamlessly. It simplifies the appointment scheduling process, allowing patients to secure their desired appointment times within the clinic efficiently.

4.1.2 Interfaces

Patient Interface: This interface enables patients to interact with the system, allowing them to book appointments and access contact information and information about us.

Dentist and Staff Interface: Dental professionals and employees can use this interface to access and manage patient records, schedule appointments, and process payments.

Processing Payments: The interface facilitates cash transactions exclusively, allowing the clinic to accept payments, issue bills, and provide change as needed.

4.2 Physical Arrangement of Devices in a Typical Network

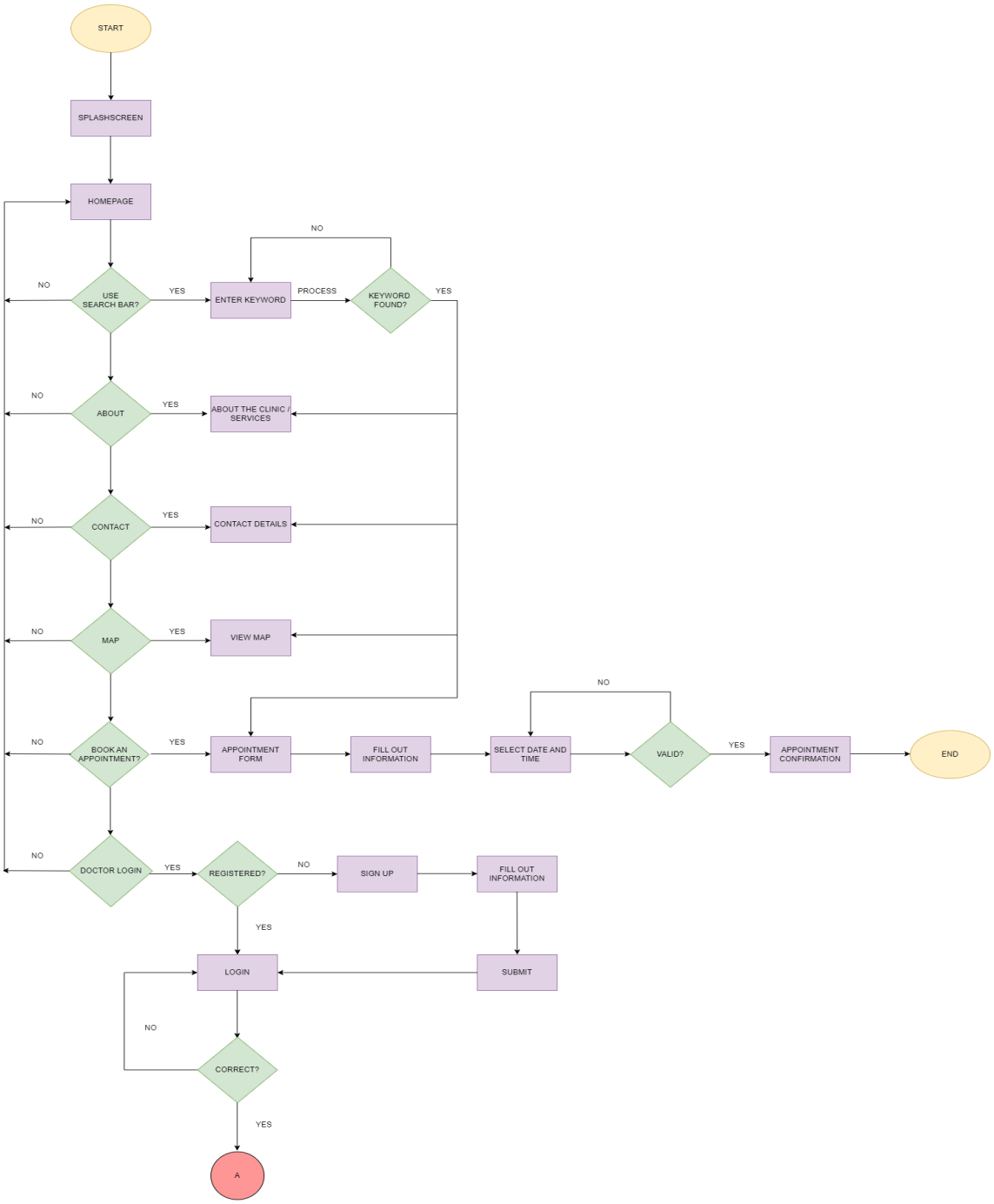
Server: The server is the main computer that contains and controls the data and software for the dental clinic system.

Workstations: Workstations are computers that dentists and staff in dental clinics use to access and communicate with the dental clinic system.

Networking Equipment: Devices like switches, routers, and firewalls that permit data transport and communication between various networked devices are included in the category of networking equipment.

Printers and Scanners: Dental clinics frequently have printers and scanners to print patient forms, treatment plans, and other paperwork.

4.3 User Flow Diagram



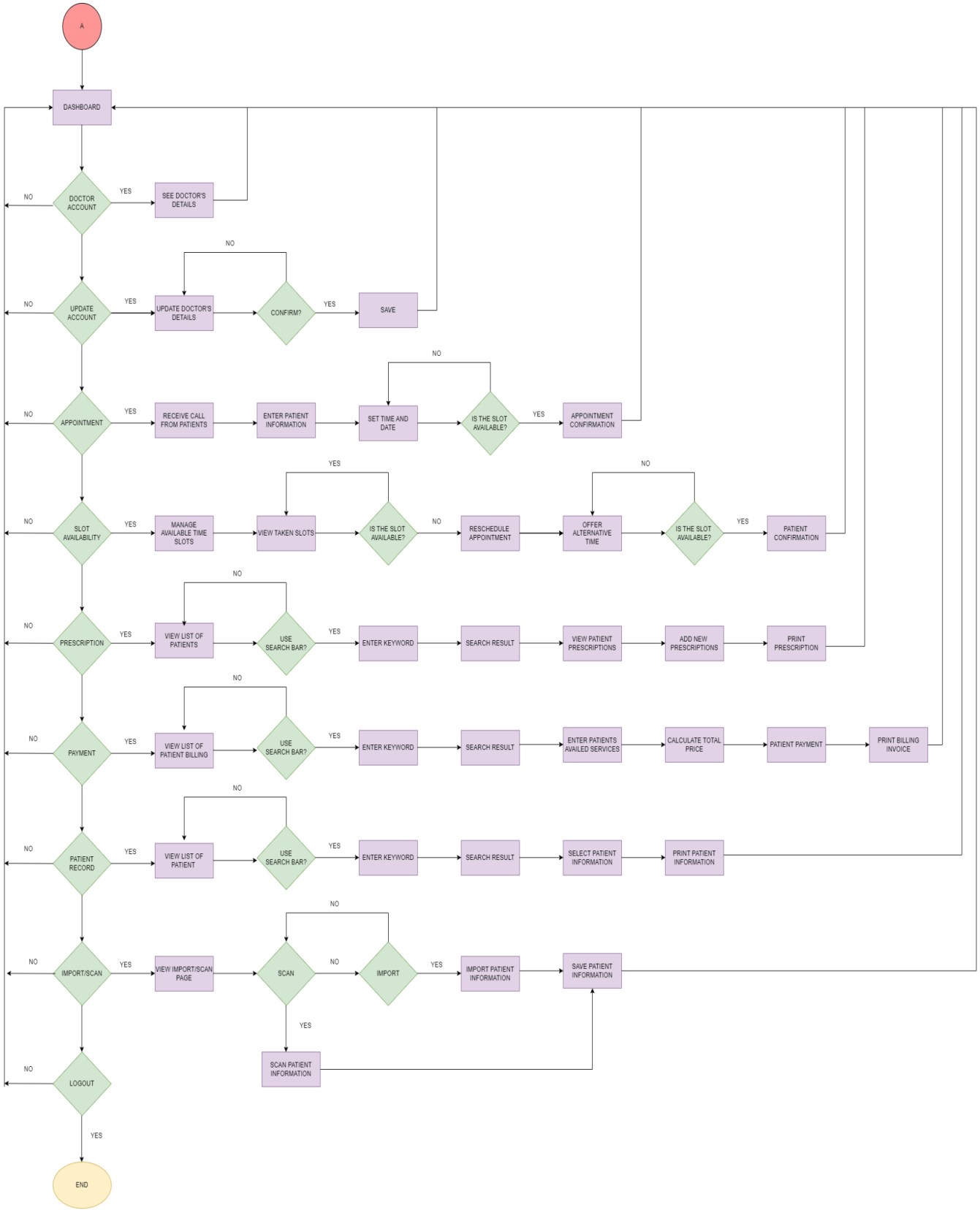

A
DASHBOARD
DOCTOR ACCOUNT
YES
NO
SEE DOCTOR'S DETAILS
UPDATE ACCOUNT
UPDATE DOCTOR'S DETAILS
CONFIRM?
SAVE
APPOINTMENT
RECEIVE CALL FROM PATIENTS
ENTER PATIENT INFORMATION
SET TIME AND DATE
IS THE SLOT AVAILABLE?
APPOINTMENT CONFIRMATION
SLOT AVAILABILITY
MANAGE AVAILABLE TIME SLOTS
VIEW TAKEN SLOTS
RESCHEDULE APPOINTMENT
OFFER ALTERNATIVE TIME
PATIENT CONFIRMATION
PRESCRIPTION
VIEW LIST OF PATIENTS
USE SEARCH BAR?
ENTER KEYWORD
SEARCH RESULT
VIEW PATIENT PRESCRIPTIONS
ADD NEW PRESCRIPTIONS
PRINT PRESCRIPTION
PAYMENT
VIEW LIST OF PATIENT BILLING
ENTER PATIENTS AVAILED SERVICES
CALCULATE TOTAL PRICE
PATIENT PAYMENT
PRINT BILLING INVOICE
PATIENT RECORD
VIEW LIST OF PATIENT
SELECT PATIENT INFORMATION
PRINT PATIENT INFORMATION
IMPORT/SCAN
VIEW IMPORT/SCAN PAGE
SCAN
IMPORT
IMPORT PATIENT INFORMATION
SAVE PATIENT INFORMATION
SCAN PATIENT INFORMATION
LOGOUT
END

FIGURE 4

This diagram has a variety of shapes, like rectangles that represent actions or processes, diamonds that represent decision points, and ovals that represent start and endpoints. The color of the shapes distinguishes the different types of actions and decisions: purple is for the user actions, green is for decisions, and yellow is for the start and end points.

4.4 Use Case Diagram

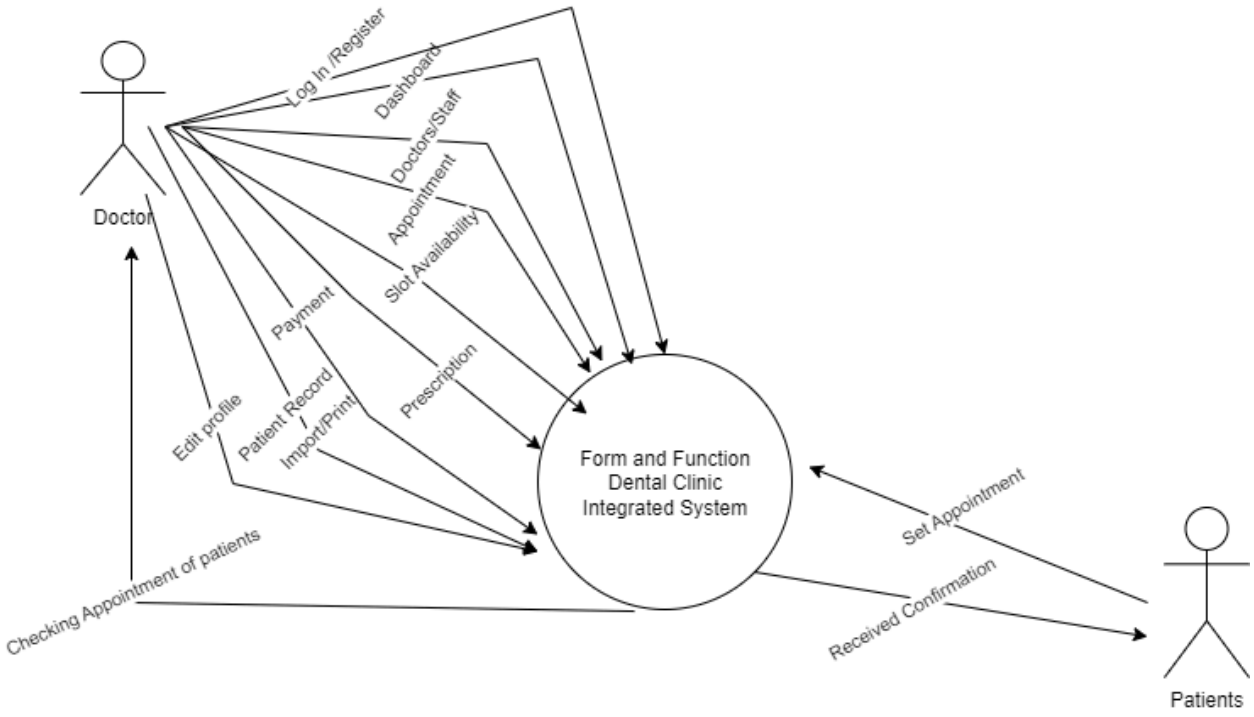


FIGURE 5

In this diagram, the doctor can perform actions in the system, like checking appointments, logging requests, and editing appointments. At the same time, The patient interaction will be to receive an email regarding the appointment schedule.

4.5 Context Diagram

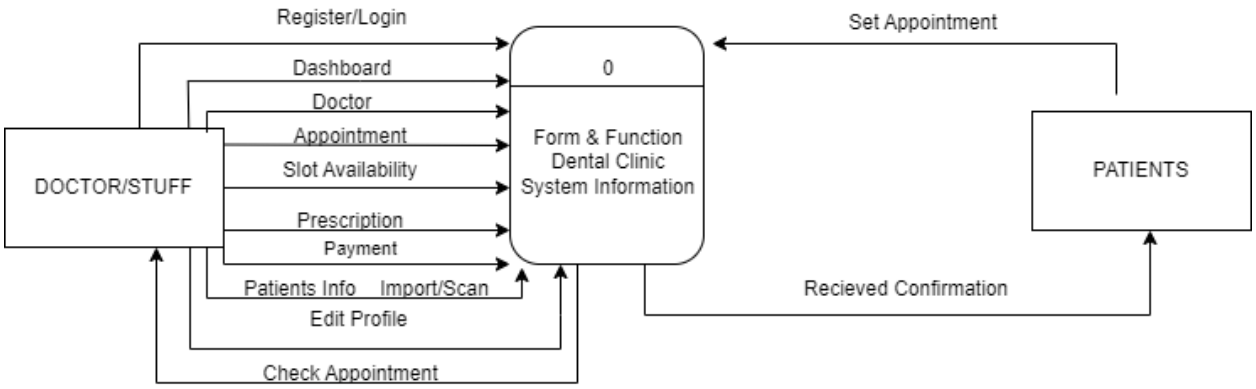


FIGURE 6

In the login/register, this box indicates the entry point to the system where doctors can log in or register new accounts. In the appointment form in that category, all of the patient's information is included in the form for the scheduling. The slot availability checks all the available slots for appointments to ensure no conflicts regarding scheduling an appointment. Lastly, the patient prescription indicates that the system can generate prescriptions.

5 Functional and Non-Functional Requirements 1

5.1 Functional Requirement

Req.ID	Description	Priority	Users
FR-001	Creation of System for Dental Clinic	Must	Doctors/Patients
FR-002	Create an account and to log in for Doctors	Must	Doctors/Staff
FR- 003	Adding a Useful “Dashboard”.	Must	Doctor/Staff
FR-004	Staff Logins & Presence Monitoring System. “Doctors”	Must	Doctor/Staff
FR-005	Walk-In “Appointments” for Accessibility.	Must	Doctor/Staff
FR-006	Real-Time “Slot Availability” Tracker.	Must	Doctor/Staff
FR-007	Online “Prescription” after the service	Must	Doctor/Staff
FR-008	Confirming “Payment”	Must	Doctor/Staff
FR-009	“Patient Record” Management	Must	Doctor/Staff
FR-0010	“Edit Profile”	Must	Doctor/Staff
FR-0011	Quality Commitment in “Home”	Must	Patients
FR-0012	Comprehensive Dental Offerings “About”	Must	Patients
FR-0013	Patients concern called “Contact us”	Must	Patients

FR-0014	“Appointment” form for patients	Must	Patients

5.1.1 Functional Requirement Description

Item	Description
Request ID	FR-001
Description	Creation of System for Dental Clinic
Priority	Must
Functional Requirement Description	● Creating a dental clinic's website showcasing a comprehensive range of dental services, offering convenient appointment scheduling and patient record management, and featuring testimonials to demonstrate the clinic's excellence in dental care. ● The system features functional buttons to manage patient records and appointments efficiently. This streamlined approach eliminates the need for manual processes, providing doctors with improved monitoring capabilities and simplifying appointment scheduling for a more seamless experience. ● Patients can utilize these systems to quickly check for available slots or their preferred schedules, providing a convenient way to determine appointments.
Acceptance Criteria	● After implementing the system, the features will showcase the dental clinic website, highlighting exclusive offers and key features for an engaging and

	informative online experience.

Item	Description
Request ID	FR-002
Description	Create an account and to log in for Doctors
Priority	Must
Functional Requirement Description	● Doctors and staff must register on the website to gain access as administrators, allowing them to log in and manage the site effectively.
Acceptance Criteria	● Upon login, doctors and staff gain access to the system featuring a dashboard, a log of logging doctors, the option for walk-in appointments, real-time slot availability, prescription management, payment, receipt generation, and a record of patients who have scheduled appointments.

Item	Description
Request ID	FR-003
Description	Adding a Useful “Dashboard”.

Priority	Must
Functional Requirement Description	● Users will find dedicated sections for Doctors, Appointments, Slot Availability, Prescription, Payment, and Patient Records within the Dashboard button, providing a convenient and organized overview.
Acceptance Criteria	● Ensuring all buttons' functionality and usability guarantees a smooth and efficient user experience.

Item	Description
Request ID	FR-004
Description	Staff Logins & Presence Monitoring System. “Doctors”
Priority	Must
Functional Requirement Description	● The Doctor's button is exclusively accessible to authorized personnel, allowing only doctors and staff to enter the website. Additionally, it provides a monitoring feature to track active users within the system.

Acceptance Criteria	● Ensuring the button's functionality by validating input data before finalizing the submission process.

Item	Description
Request ID	FR-005
Description	Walk-In “Appointments” for Accessibility.
Priority	Must
Functional Requirement Description	● Walk-in appointments cater to individuals who may not be familiar with the online appointment process, offering a convenient solution for the elderly, persons with disabilities, and others who may find it challenging to schedule appointments online.
Acceptance Criteria	● The website prioritizes data security and privacy while still aiming to provide a user-friendly experience for patients. To ensure confidentiality and ease of use, the submission process includes measures such as thorough input data validation.

Item	Description
Request ID	FR-006
Description	Real-Time “Slot Availability” Tracker.
Priority	Must
Functional Requirement Description	● The slot availability feature not only displays open schedules but also monitors the scheduled appointments, providing a comprehensive overview of patient and doctor availability at specific times.
Acceptance Criteria	● Guaranteeing transparent slot availability, our system monitors appointments, ensuring a comprehensive patient and doctor availability overview. Rigorous data validation measures are in place to prioritize accuracy and privacy, enhancing the security of Patients.

Item	Description
Request ID	FR-007
Description	Online “Prescription” after the service
Priority	Must

Functional Requirement Description	● In the Prescription section, detailed patient information, including name, address, gender, and date of birth, is displayed. Additionally, it includes a comprehensive description of the prescribed medication or treatment provided by the doctor, creating a link between the prescribed service and the patient's specific health needs.
Acceptance Criteria	1. All information about us must be created and existed in the Form & Function Dental Clinic 2. All mandatory fields must be input and checked for validity before the Form & Function Dental Clinic is created.

Item	Description
Request ID	FR-008
Description	Confirming “Payment”
Priority	Must
Functional Requirement Description	● In the Payment section, after receiving the service, patients can make payments. This includes the payment time, contact

	number, date, and patient name for a straightforward and efficient process.
Acceptance Criteria	1. The Form & Function Dental Clinic should contain all relevant information about us. 2. Ensure all required fields are filled in and validated before creating the Form & Function Dental Clinic.

Item	Description
Request ID	FR-009
Description	“Patient Record” Management
Priority	Must
Functional Requirement Description	● The Patient Record section provides comprehensive information about individuals who book appointments at our clinic. This feature makes printing and archiving records easy, resulting in a well-maintained and easily accessible patient history within the clinic.
Acceptance Criteria	● The Patient Record section is designed to maintain meticulous records while ensuring the utmost privacy and security of an individual's details. This feature provides an organized repository of patient histories within the clinic, including the ability to print and archive records. To ensure accuracy, strict measures are in place to validate data. The result is a comprehensive and secure patient record-keeping process.s.

Item	Description
Request ID	FR-0010
Description	“Edit Profile”
Priority	Must
Functional Requirement Description	● Doctors and staff can correct their information in the Edit Profile section, ensuring accuracy. This feature enables the seamless saving and updating of their profiles with the provided information.
Acceptance Criteria	1. Include all pertinent details about us within the Form & Function Dental Clinic. 2. Verify the completion and validity of all mandatory fields before establishing the Form & Function Dental Clinic.

Item	Description
Request ID	FR-0011
Description	Quality Commitment in “Home”
Priority	Must
Functional Requirement Description	● This button directs users to the Service offering section on the home page, where the clinic details the qualifications and credentials of its staff, including licensed professionals. This information is provided to establish clear expectations for customers.
Acceptance Criteria	1. Ensuring transparency and customer

	confidence, the clinic meticulously outlines the qualifications and credentials of its staff, emphasizing licensed professionals, to set clear and trustworthy expectations for customers. 2. The Form & Function Dental Clinic should contain all relevant information about us. 3. Ensure that all required fields are filled in and validated before creating the Form & Function Dental Clinic.

Item	Description
Request ID	FR-0012
Description	Comprehensive Dental Offerings “About”
Priority	Must
Functional Requirement Description	● In the 'About Us' section, from the patient's perspective, highlight the array of services offered by the clinic.
Acceptance Criteria	● Ensuring a patient-centric approach in the 'About Us' section showcases the diverse range of services available at the clinic.

Item	Description
Request ID	FR-0013
Description	Patients concern called “Contact us”
Priority	Must

Functional Requirement Description	● In the 'Contact Us' section, find assistance for concerns by accessing the provided contact number, email, and Address. Additionally, obtain information about the clinic's location if needed.
Acceptance Criteria	1. Ensure all relevant information about us is incorporated into the Form & Function Dental Clinic. 2. Validate the completeness and accuracy of mandatory fields before creating the Form & Function Dental Clinic.

Item	Description
Request ID	FR-0014
Description	“Appointment” form for patients
Priority	Must
Functional Requirement Description	● "In the 'Appointment' section, from the patient's perspective, facilitate the scheduling process by allowing them to input their availability and personal information, time, and date preferences..
Acceptance Criteria	1. Ensuring a patient-friendly experience, the 'Appointment' section allows individuals to seamlessly schedule by providing their availability, personal details, and preferred time and date. 2. The Form & Function Dental Clinic should contain all relevant information about us. 3. Ensure all required fields are filled in and validated before creating the Form & Function Dental Clinic.

5.2 Non-Functional Requirements

Requirement ID	Description	Priority	Users
NFR-001	System Audit	Must	Doctor
NFR -002	System Control	Must	Doctor
NFR -003	System Security	Must	Doctor
NFR -004	Backup and Disaster Recovery	Must	Doctor
NFR -005	Usability Requirements	Must	Doctor

5.2.1 System Audit

Example:

The System shall provide audit trail functionalities which include:

1. Keep a record of user login/logout information, e.g., also can update the information of the doctor.
2. Record the patients' appointments by filling out the (Name, Address, Email, DOB, Gender, Schedule, and Contact no.)
3. Keep a record of the rescheduling of the patients.
4. Keep a record of the Prescription such as (Name, Date, Medication, Amount, and Frequency)
5. Keep a record of the Payment.
6. Keep a record of the patient info's scan/import (print).

5.2.2 System Control

The System shall provide Control functionalities which include:

1. Doctors must make an account and log in to the system.
2. Patient groups are created according to the level of access through appointment forms, slot availability, and rescheduling.
3. Integrated System administration functions should be defined in a separate function dashboard and separated from other normal patient functions.

5.2.3 System Security

The System shall provide Security functionalities which include:

1. All Doctors have Password protection and strong passwords for doctor accounts.
2. All imported/uploaded data files must be virus-free.

5.2.4 Backup and Disaster Recovery

Example:

The System shall provide Backup and Disaster Recovery functionalities which include:

1. The system shall provide Regular data backups by taking a snapshot of the database at specific time intervals (e.g., daily, weekly).

5.2.5 Usability Requirement

The system should meet the following usability requirements:

1. The Dashboard should be clever and consistent throughout the system.
2. Ensuring that the system is responsive and works seamlessly on Windows 10, 11, and IOS, Desktop, Laptop, and Android
3. Display available time slots clearly through the calendar so they can choose from the options that suit their time.

5.2.6 System Availability

1. The system must provide service with a high availability of 99.5% during regular office hours (i.e., from 9:00 a.m. to 6:00 p.m. Monday to Sunday).
2. Maintenance can be done any time after regular office hours.

5.2.7 System Performance

Ensuring the website's performance is important for the users' experience. It shall provide the following:

1. Loading Page Speed: Optimizing images and formats to reduce loading time.

2. Optimize server configurations for faster loading time.

3. Test the website on different browsers to ensure optimal performance

4. Regularly cleaning of databases to improve database performance.

5. The website should handle a significant number of users without dropping the performance speed of the website

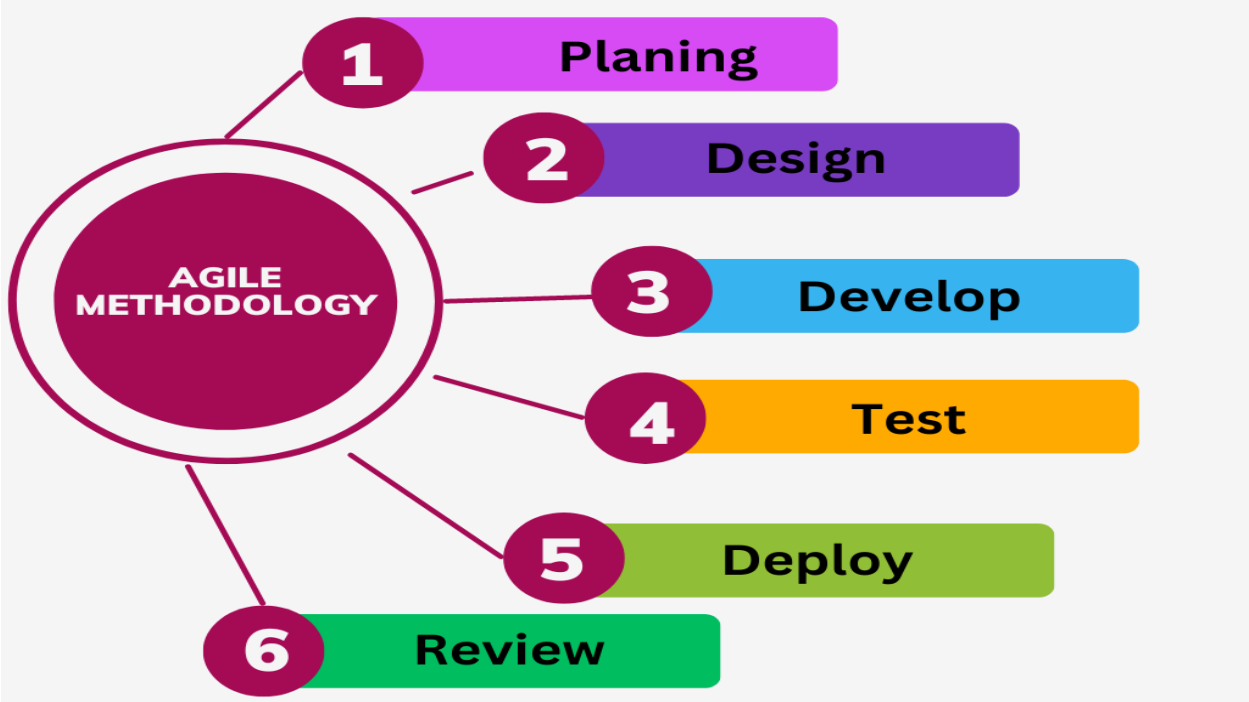
5.2.8 Number of Users and IT Equipment Requirement

- **Number of User**s: The company's size and customers will determine the number of users. The users can be identified as patients or doctors.

- **IT Equipment Requirement**: Implementing this project would require computers mobile devices to ensure responsiveness and user experience, web servers, networking equipment such as routers, switches, LAN cables, database management software like MySql, etc.

6 Software Model and Design

6.1 Software Development Model



The agile model has a crucial first phase called Planning, which involves defining objectives and requirements. In this case, the Form and Functional Dental Clinic is the target audience for the integrated system. A prioritized list of features to be developed is needed. Moving on to the next phase, we have The Design. This part is where the conceptual and visual aspects of the Form and Function of the Dental Clinic Integrated System are created. We aim to design the user interface (UI) as intuitive, user-friendly, and visually appealing, aligning with the clinic's branding and overall identity. The third phase is Development. It includes creating appointment booking forms, Patient Records, and service pages while applying coding practices and adhering to web standards for optimal performance and security. After that, we have the test phase. This part is crucial for ensuring the quality of the Form and Functional Dental Clinic Integrated System. Testing helps us identify and resolve issues before deployment of the system. Once the system passes the testing process, it is ready for deployment, the fifth stage. Lastly, the Review phase involves collecting feedback from the doctors, staff, and patients to assess the system's

effectiveness and gather suggestions for improvements. Feedback is gathered by asking the opinion of the doctor, staff, and patients.

6.2 Sequence Diagram

6.2.1 Sequence Diagram 01: Patient

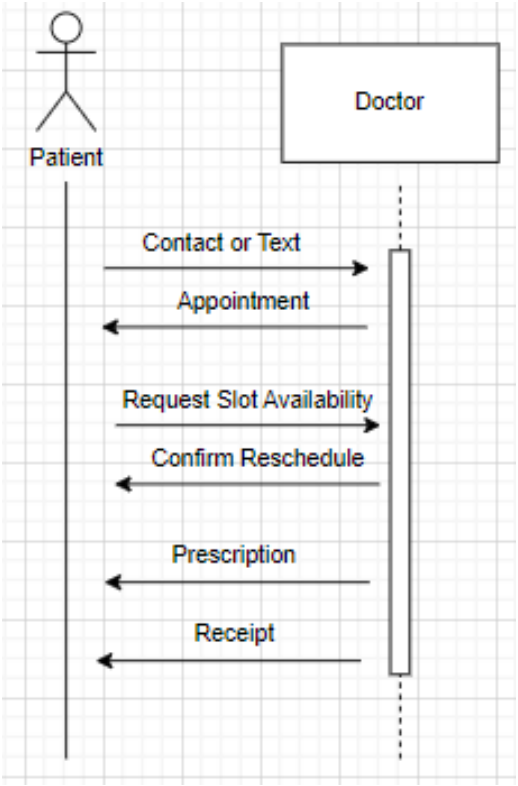


FIGURE 7

In this diagram, we use arrows to show the directions of communication and exchange of information between the patient and the doctor. The straight vertical line suggests a sequential process flow the doctor follows in response to the patient's actions.

6.2.2 Sequence Diagram 02: Doctor

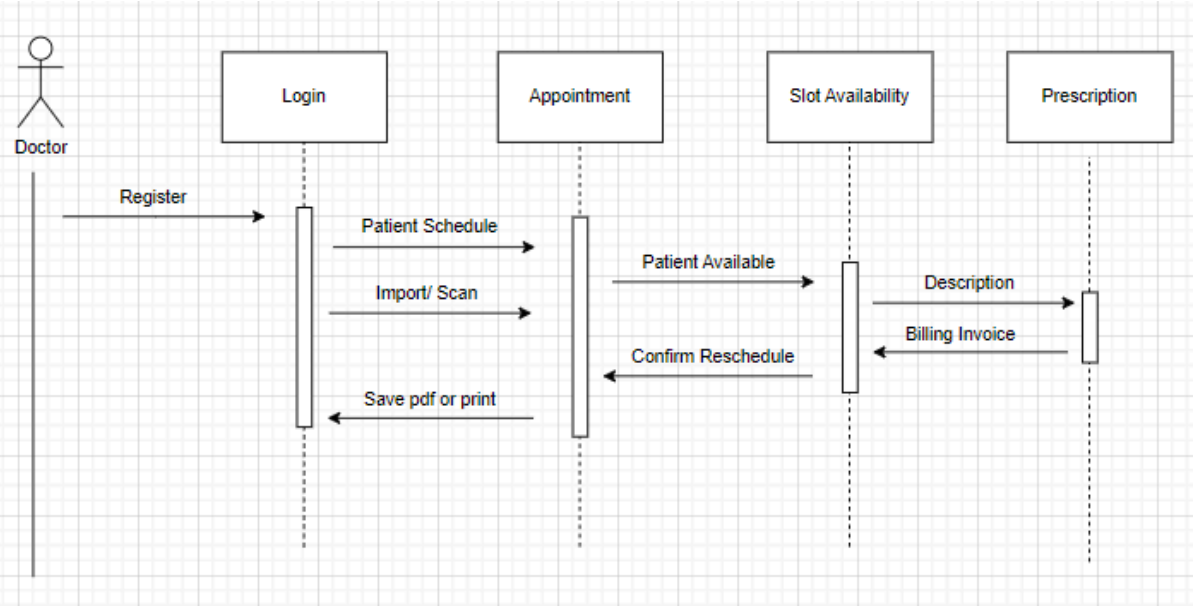


FIGURE 8

In this diagram, the first step involves the doctor registering to access the system. After successfully registering, the doctor will be able to login and gain access to the system. The system will help the doctor follow the patient's schedule, which includes importing and scanning documents related to the patient. The doctor will then be able to schedule appointments for the patients, check for their availability, and confirm rescheduling if necessary. Finally, the doctor can prescribe medication and generate patient billing invoices.

6.3 ER Diagram

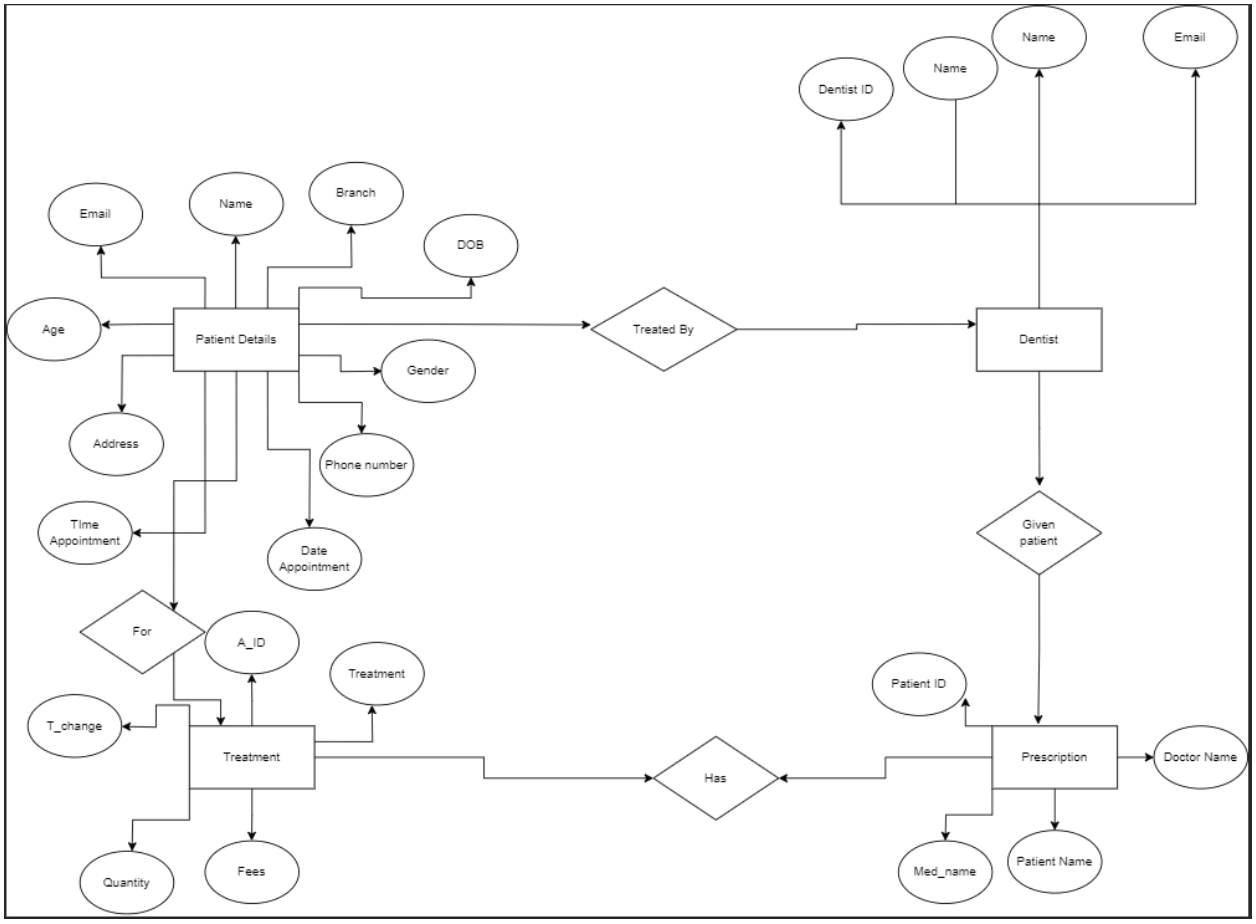


FIGURE 9

The login and register functions are linked to the doctor's login ID and password. Additionally, the email is connected to the login ID, while the Appointment ID is linked to both the email and the patient's schedule. The patient's schedule, in turn, is tied to the appointment ID and appointment. The appointment branches out to the outpatient schedule, prescription for the patient, slot availability, and confirmation or rescheduling options.

6.4 Database Schema

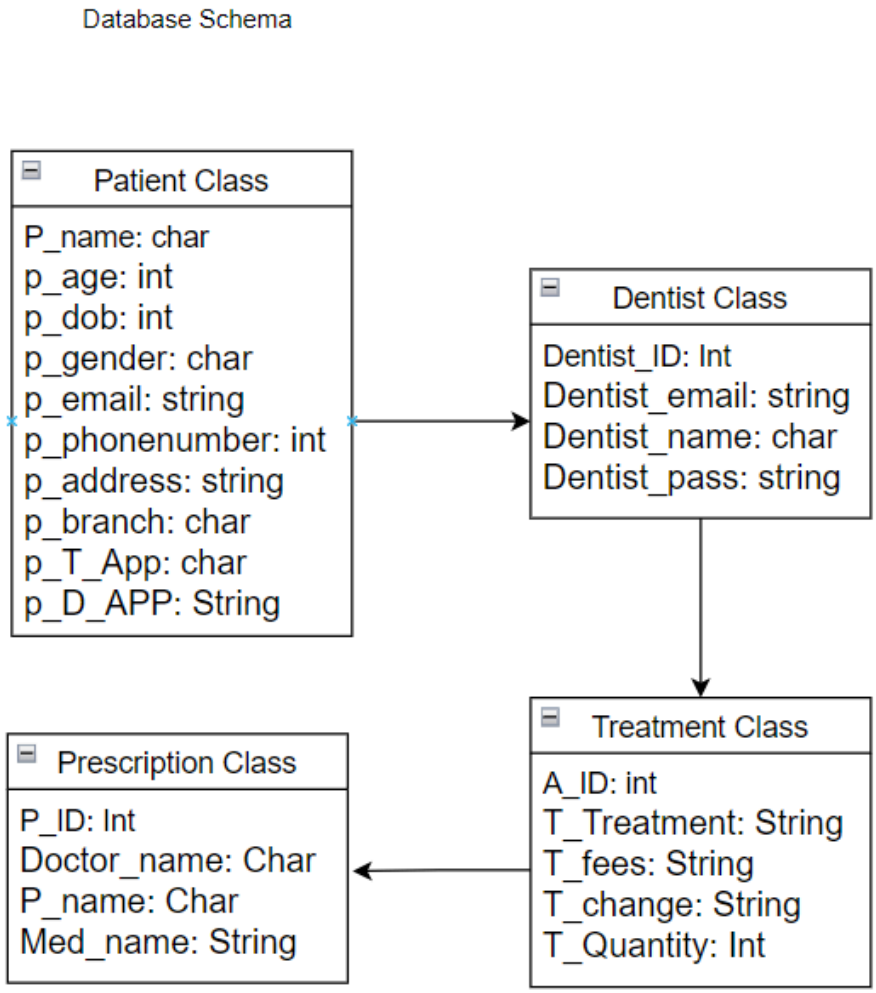


FIGURE 10

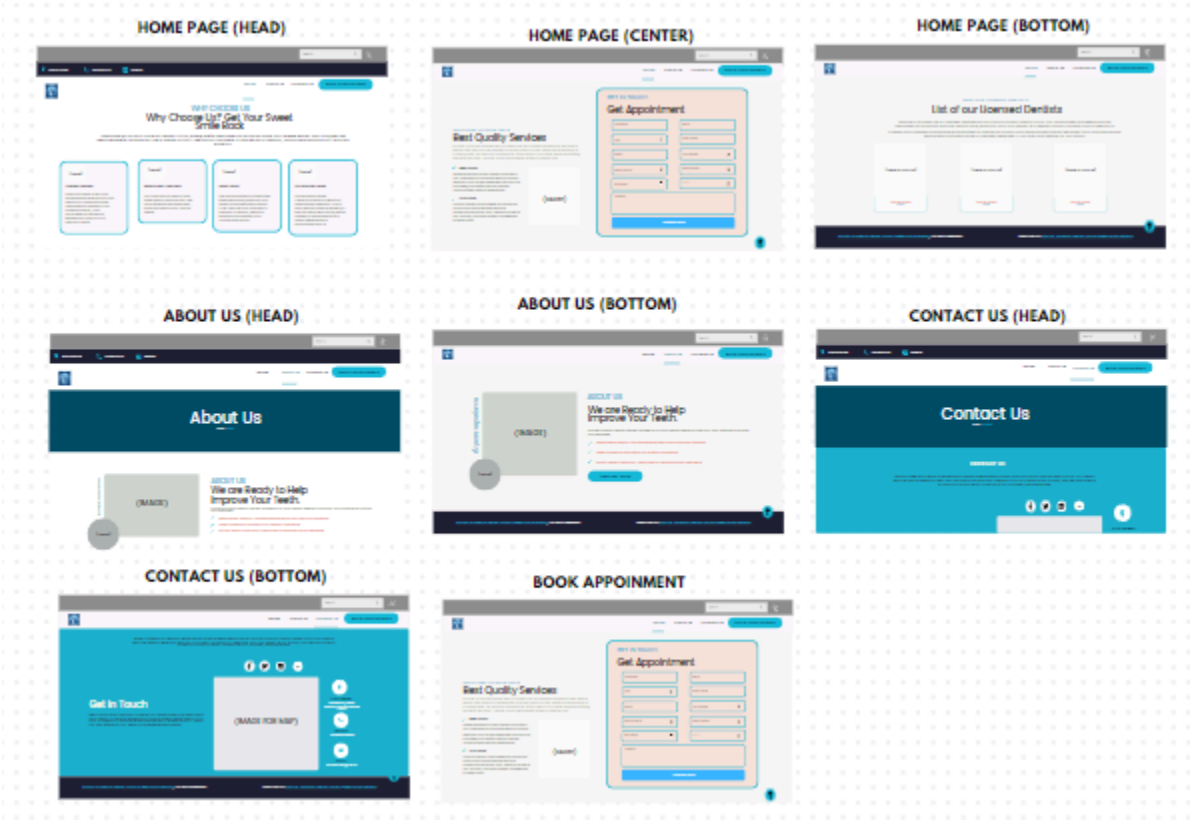
The PK in this model refers to the primary key for each entity. The "sign up doctor" and "login doctor" entities represent the processes of a doctor registering and logging into the system, respectively. These entities contain attributes such as login ID, password, and other personal details. The "doctor staff" entity is almost identical to the "doctor" entity as it also has a unique and specific ID, name, and other attributes. The "appointment form" entity holds information related to patient appointments,

while the "patient record" entity contains unique patient information such as ID, DOB, gender, and contact information.

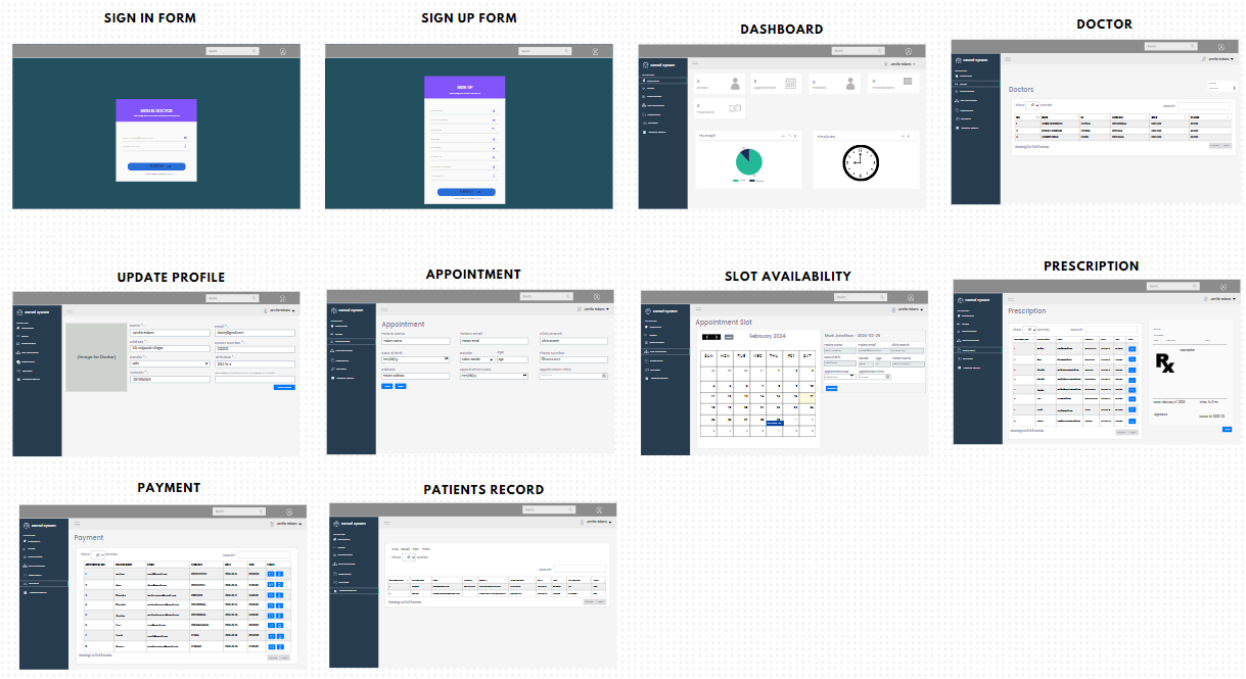
6.5 User Interface Design

6.5.1 User Interface Design: Website

Patient



Doctor / Staff



6.5.1 Rules and Guidelines for User Interface Designing

1. **Consistency**: Keep all the website's pages in the same design. To provide consumers with a seamless and recognizable experience, utilize the same color scheme, typeface, and button styles throughout the interface. Keep the website's design similar on all of its pages. To provide consumers with a consistent and recognizable experience, employ the same color scheme, typeface, and button styles throughout the interface.
2. **Simplicity**: Keep the user interface straightforward and simple to use. Make important information stand out while keeping the interface clear of extraneous items.
3. **Visual Hierarchy:** UI elements should be arranged to direct users' attention. To ensure consumers can identify and use key capabilities easily, emphasize critical activities like "Book Appointment" and "Services."
4. **Whitespace:** Leave enough whitespace between elements to make an interface clear and tidy. To improve readability and usability, keep the pages from being too crowded.
5. **Straightforward Navigation:** Create an easy-to-use navigation menu so users can easily navigate the various website parts. Label menu items clearly by using names like "Home," "Services," "About Us," and "Contact."

6. **Feedback and Error Handling:** When users complete actions, like successfully scheduling an appointment or submitting a form, provide them prompt feedback. Clearly labeled error warnings should there be any input problems.
7. **Responsive Design:** Ensure the website is responsive and can be viewed on various devices, such as computers, tablets, and smartphones. Adapt the design and information display to various screen sizes.
8. **Accessibility:** Put accessibility elements in place to ensure that all users, including those with disabilities, can use the website. Ensure photos have alt text for screen readers and that text may be resized.
9. **Usability Testing:** Real people should be used to find any interface problems or potential improvements. Utilize the comments to improve the UI incrementally.
10. **Familiar Patterns:** Make it simpler for users to connect with the website by incorporating well-known UI patterns, such as a search bar in the header, appointment booking forms, and contact forms.
11. **Minimize Cognitive Load:** Reduce cognitive strain by breaking the appointment booking procedure into straightforward phases and guiding consumers through each.
12. **Color and Contrast:** Pick a color scheme that is visually appealing and has enough contrast to allow for easy reading. To make buttons and other critical components stand out, use contrasting colors.
13. **Progress Indicators:** Provide progress indicators to inform consumers where they are in the process if there are numerous steps in the appointment booking procedure.

6.5.1.2 User Input Validation Methods

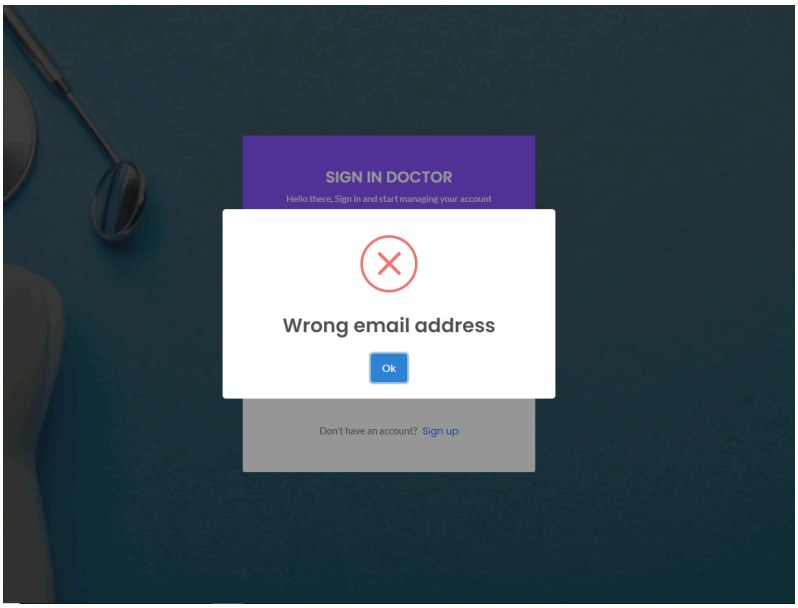
This website relied on PHP and XAMPP for its backend infrastructure, including the database management system. While there wasn't a dedicated client-side validation mechanism due to the system being exclusive to doctors within the clinic, PHP scripts handled input validation and data integrity checks on the server-side.

For instance, when doctors entered their email addresses, PHP scripts validated the input to ensure it contained the "@" symbol and at least one dot (.), with the "@" not being the first character, and the last dot appearing after the "@" sign with a minimum of two characters before the end.

By performing these validations on the server-side, the system ensured that only accurate and desired information was processed and stored securely in the XAMPP-managed MySQL database. This approach maintained data integrity and reliability within the clinic's system, supporting efficient and secure management of patient information by doctors.

6.5.1.2 Error Message, Warnings and Supportive Information

The system employed concise and clear error messages to promptly communicate any encountered issues to users during various operations. Each error message began with a clear indication of the error type, facilitating users' quick understanding of the underlying problem.



6.5.1.3 Guideline for Interface Designing

Flexible Input Methods: Ensure that all features and functionalities within our system can be accessed and operated seamlessly using either the keyboard or mouse. Users should have the flexibility to navigate through menus, select options, and interact with content without being restricted to a specific input method.

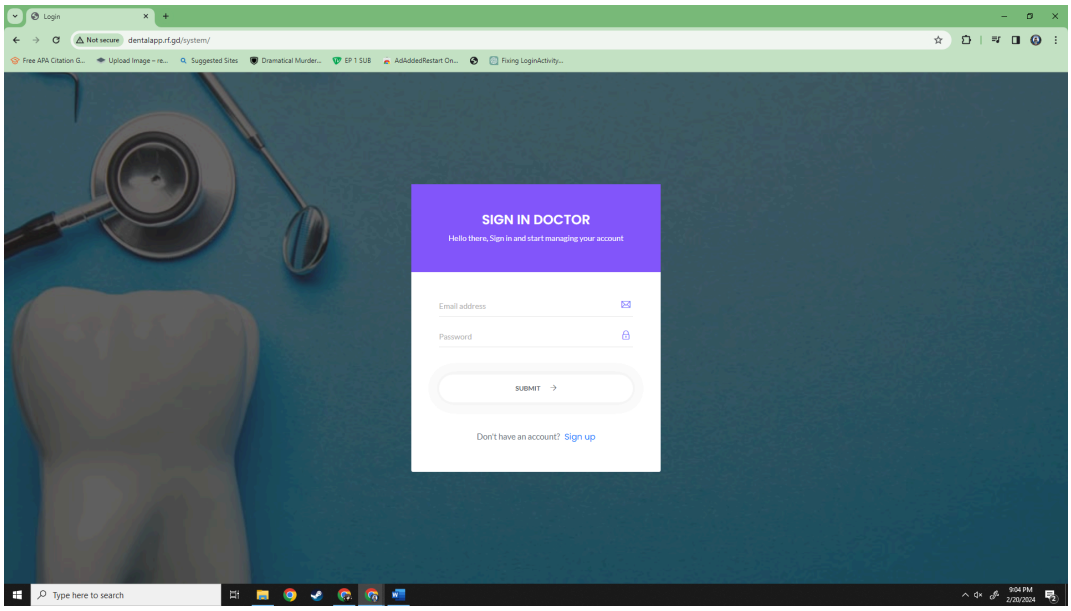
Mobile View Optimization: Ensure that the user interface is responsive and optimized for mobile devices, providing an intuitive and user-friendly experience on screens of various sizes. Utilize responsive design principles to adapt layout, font sizes, and interactive elements for optimal viewing and interaction on mobile devices. Prioritize essential features and streamline the user interface to enhance usability on smaller screens.

Clear Navigation Paths and Exits: Develop intuitive and well-defined navigation routes between various interfaces, allowing users to effortlessly locate relevant sections. Implement prominent buttons or links for quick access to the main dashboard or home page, ensuring seamless navigation throughout the system.

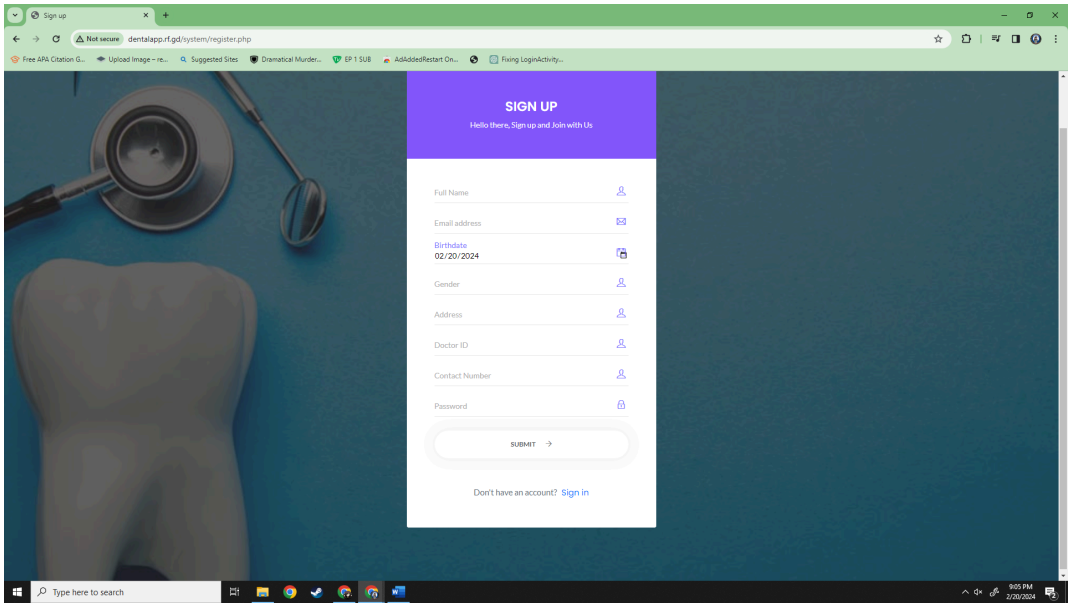
User-Friendly Interface for All Skill Levels: Craft user interfaces that are straightforward and easy to use, accommodating users with diverse technical proficiencies. Incorporate tooltips and contextual help features to provide guidance and assistance to users as they navigate through tasks and processes, enhancing usability and ensuring a positive user experience across all skill levels.

6.5.2 User Interfaces for Each Use Case

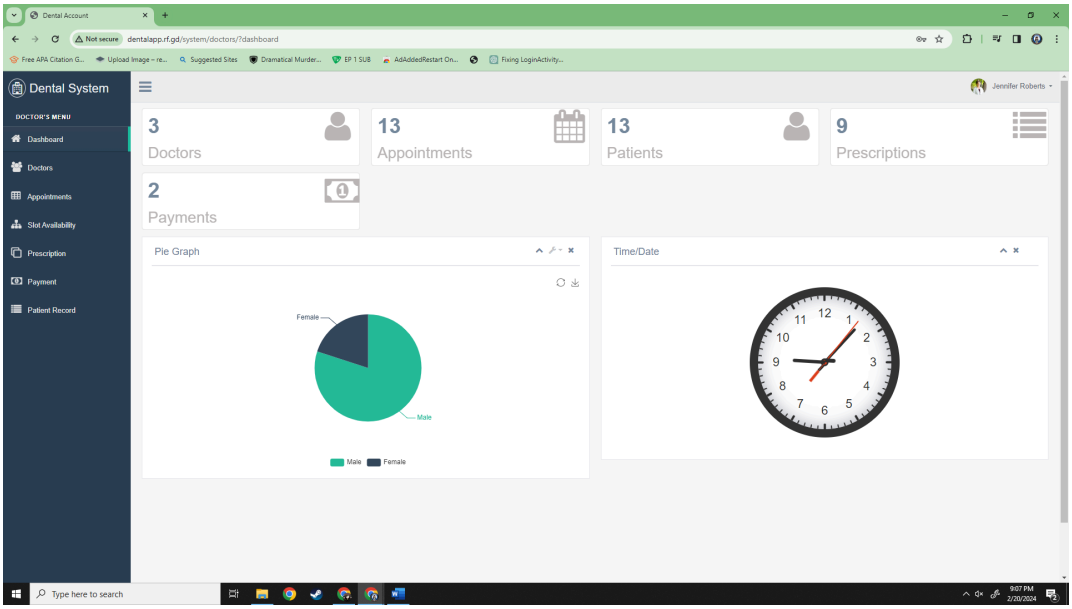
Use case 1 : LogIn into the system



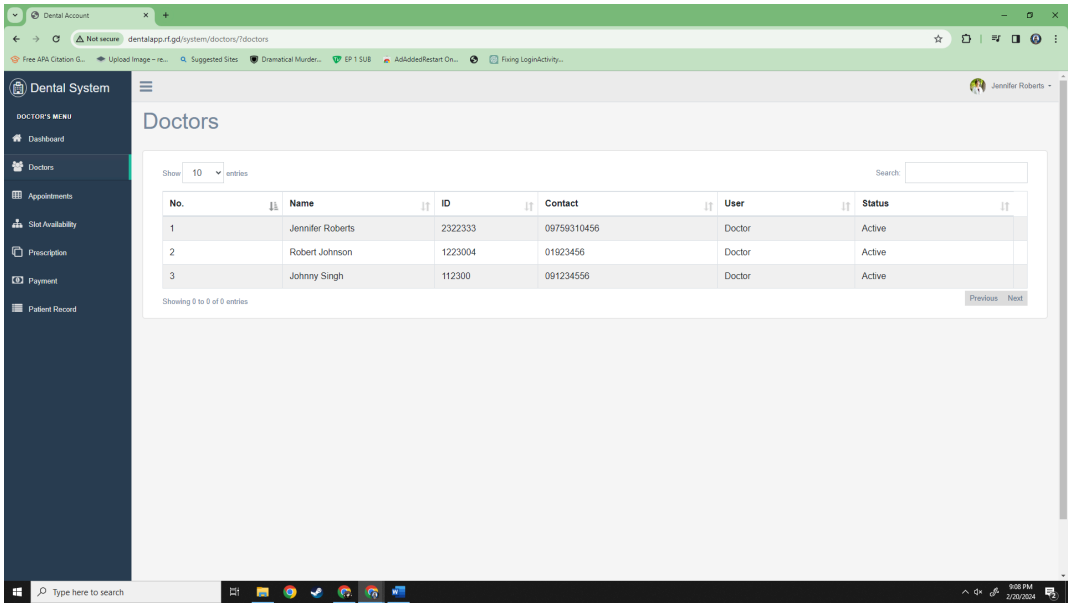
Use case 2 : Signing Up / Creating account



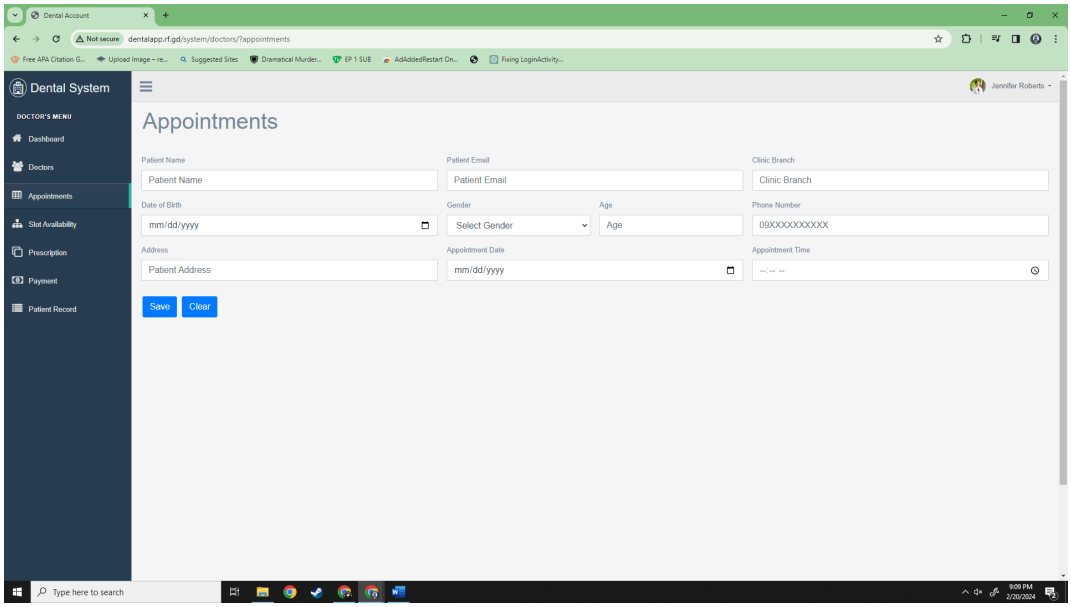
Use case 3 : Doctor Dashboard



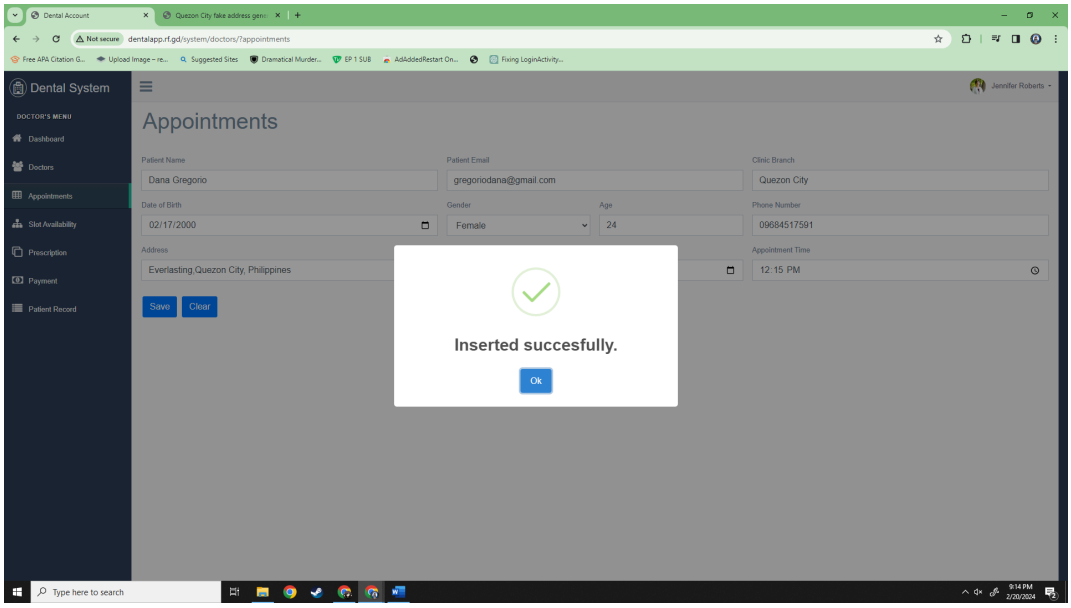
Use case 4 : List of Doctors



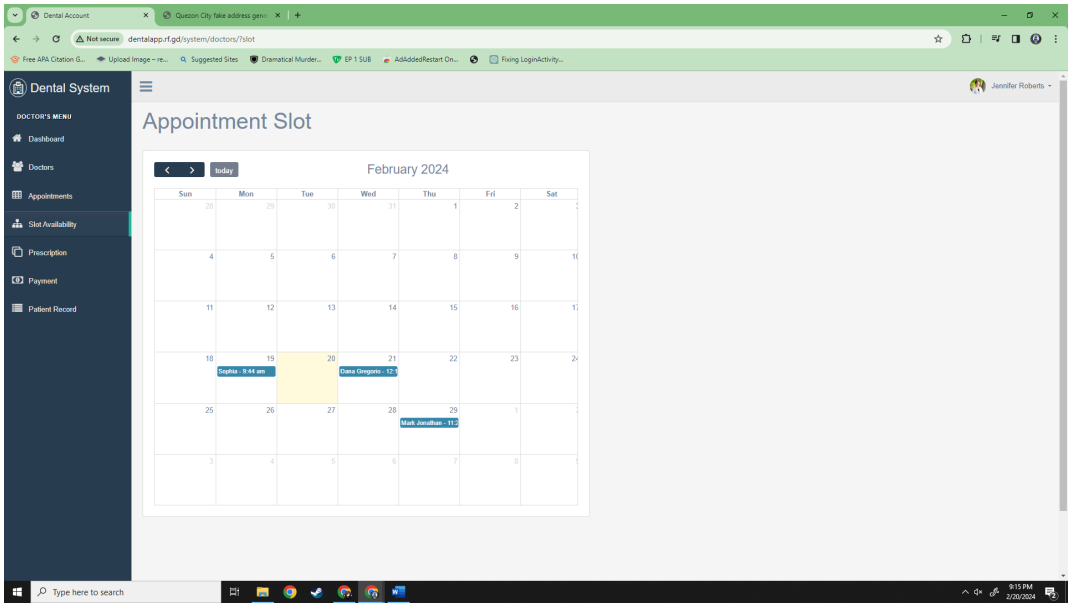
Use case 5: Create Appointment



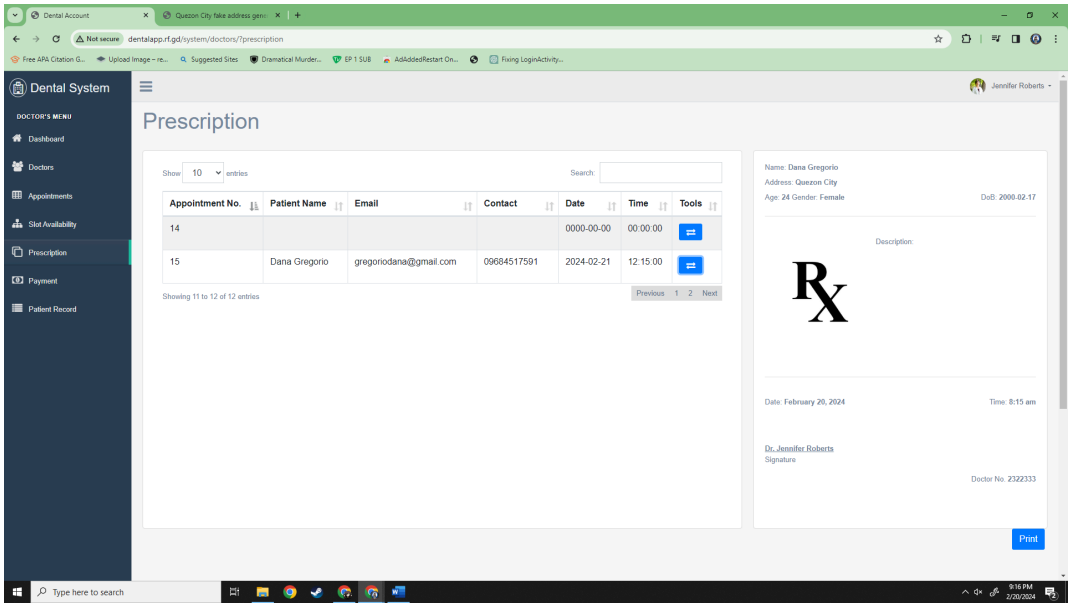
Use case 6 : Appointment Successful



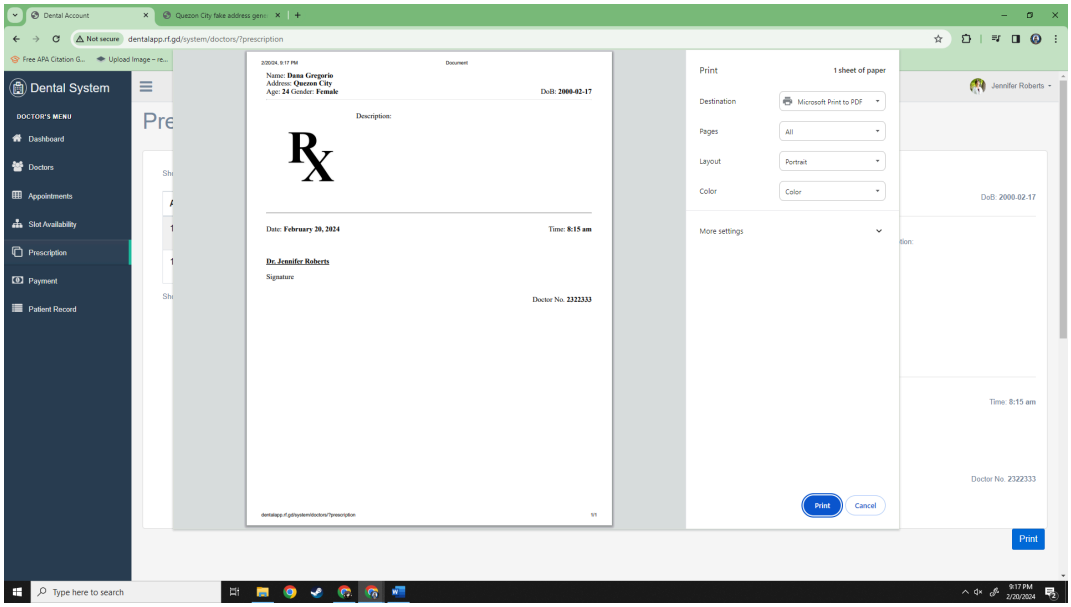
Use case 7 : Manage Appointment Slot Availability



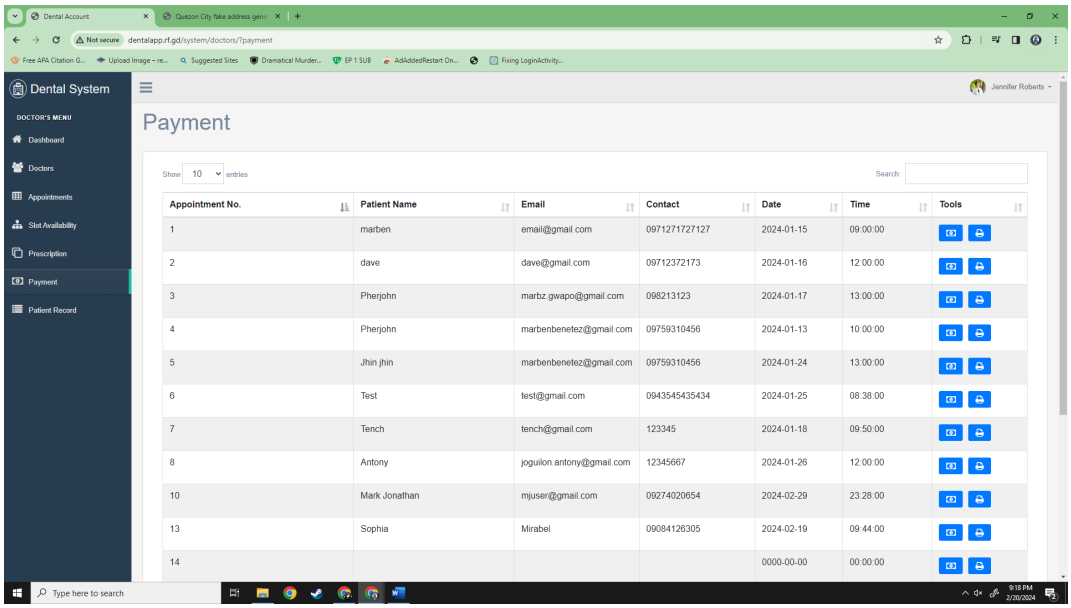
Use case 8 :Manage Prescription



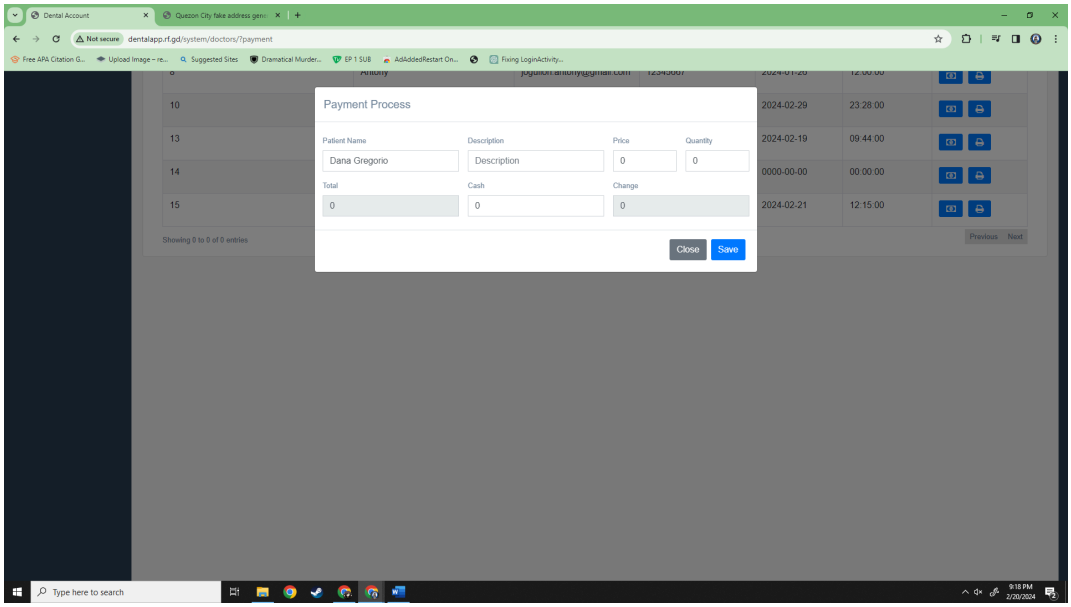
Use case 9 : Prescription Print Setup



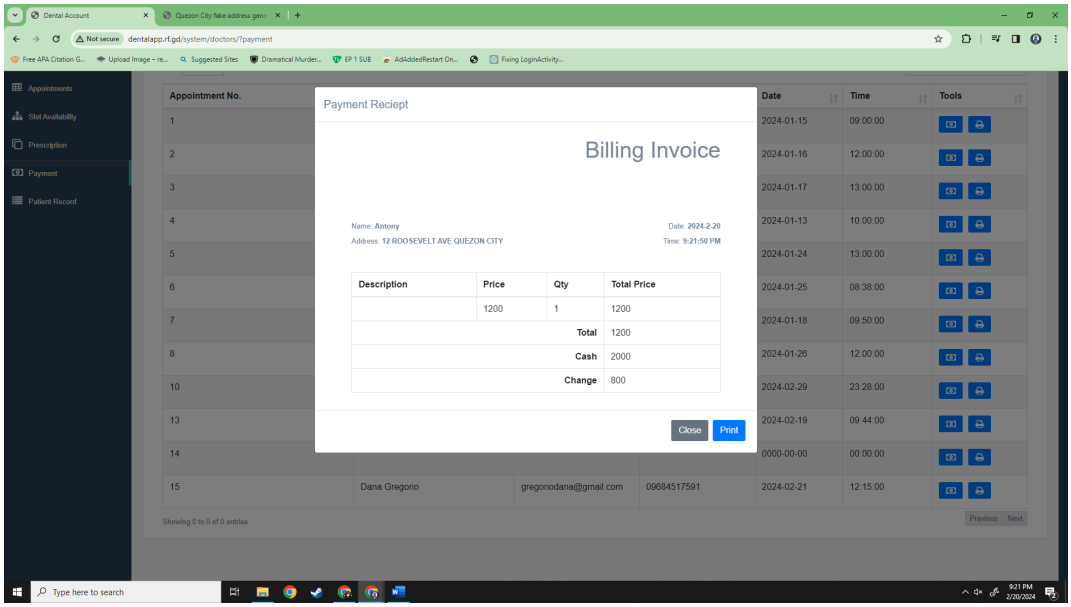
Use case 10 : Manage Payment



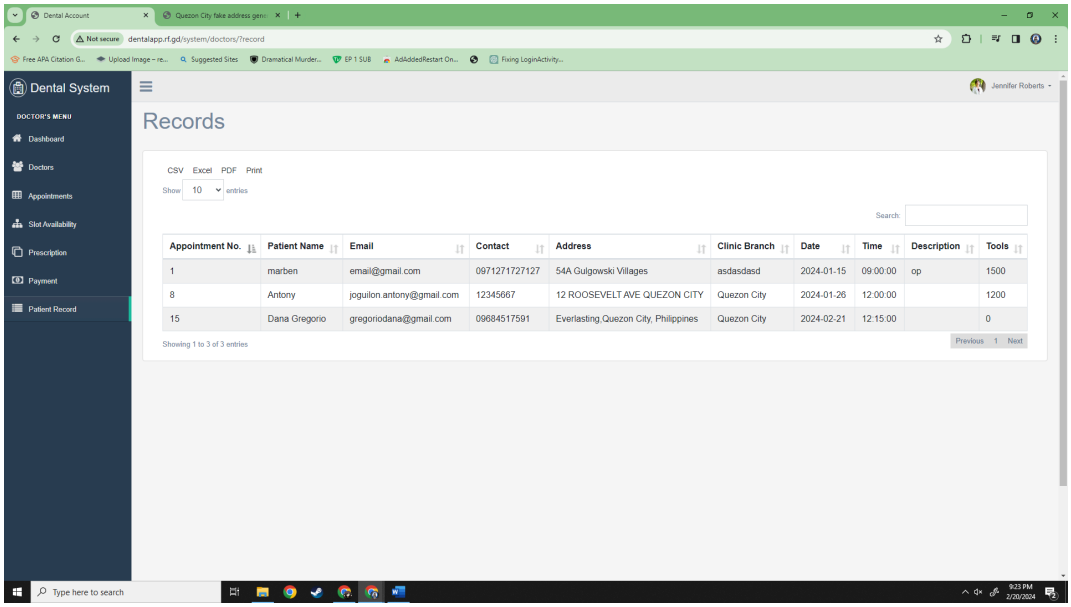
Use case 11 : Manage Payment Process



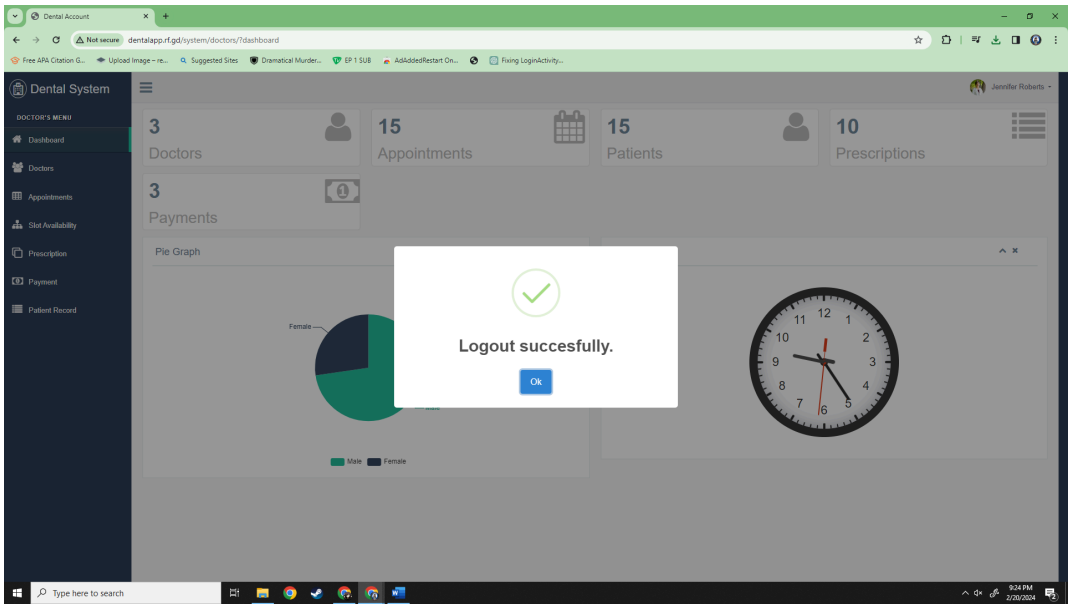
Use case 12 : Printable Receipt



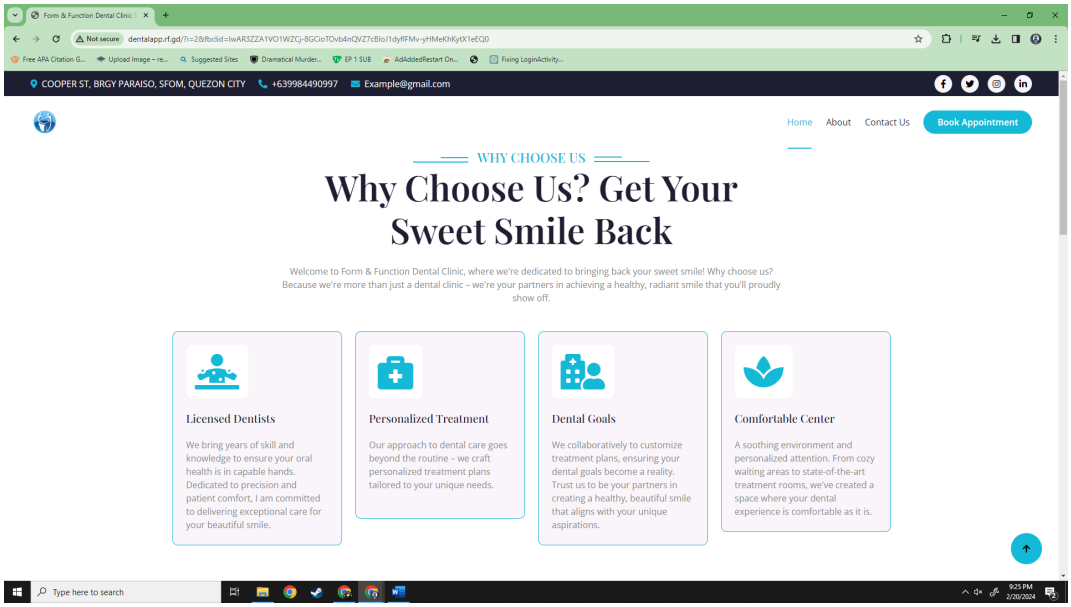
Use case 13 : Manage Patient Records



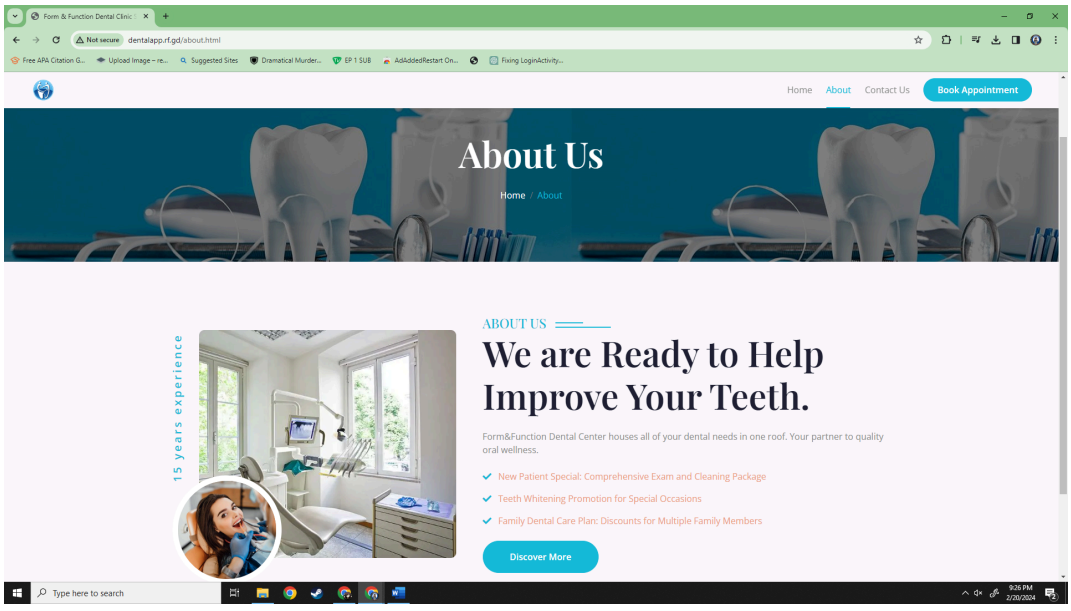
Use case 14 : Logout Success



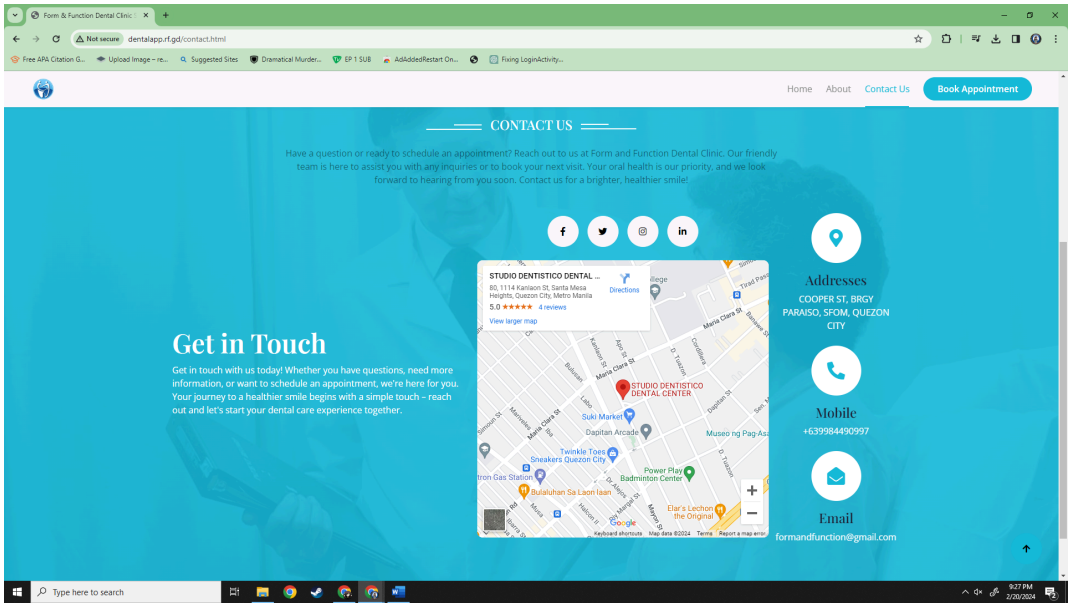
Use case 16 : Patient Homepage



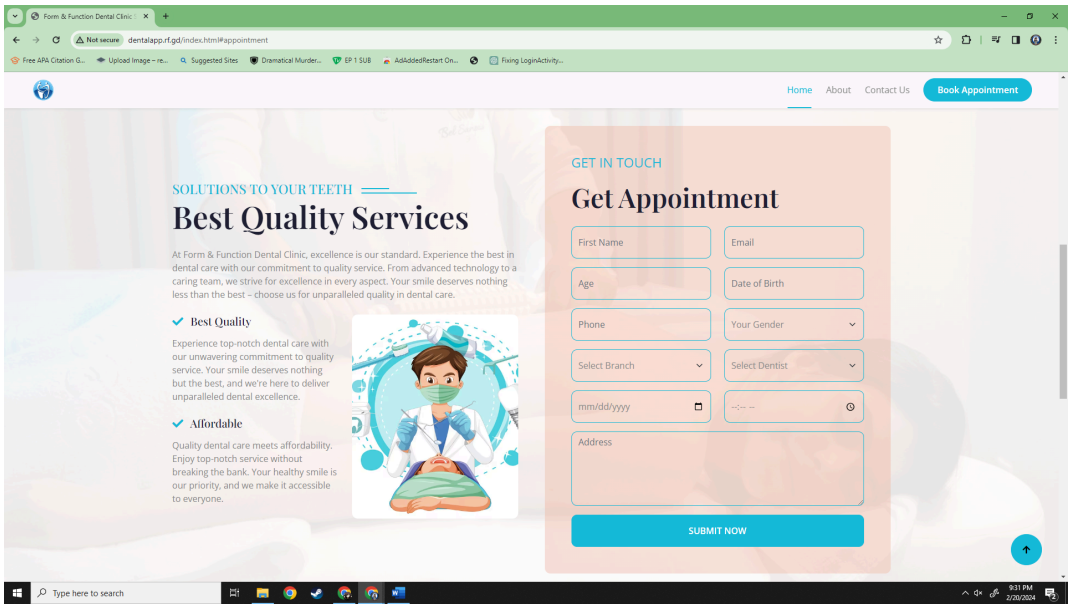
Use case 17 : About Us Page



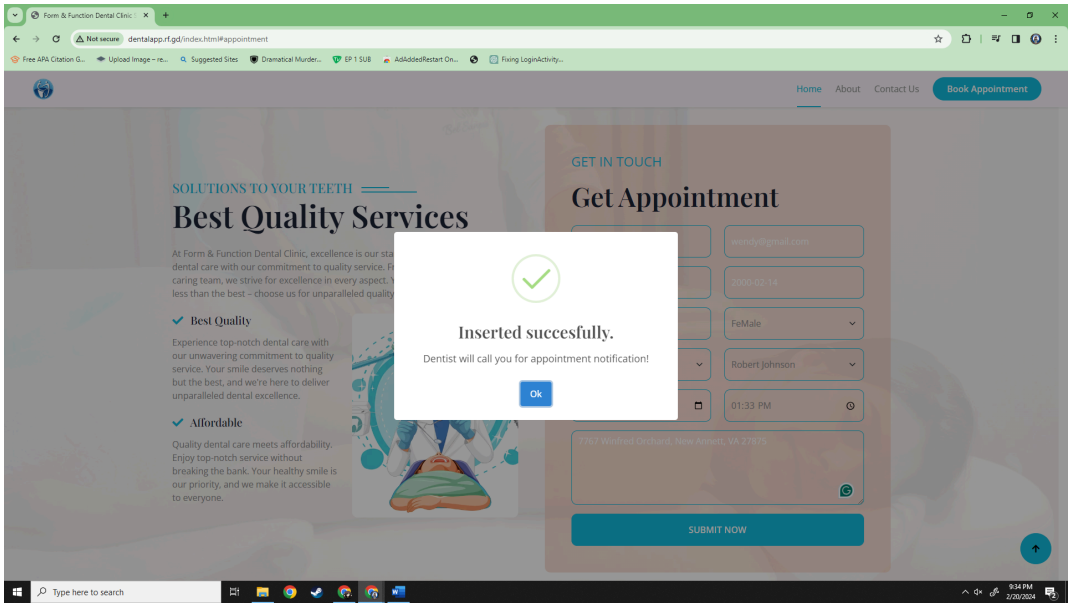
Use case 18 : Contact Us Page



Use case 19 : Appointment Form



Use case 20 : Successful Appointment Booking



User Acceptance Test Customer Page

PATIENT PAGE

S#	Perquisite
1	Desktop/Laptop
2	Internet/Wifi
3	Browser/link

Test Case ID:	TC_01

Test Scenario:	Quality Commitment in “Home”

Step #	Step Details	Expected Results	Actual Results	Remarks
1	View the quality and services that dental clinics offer.	You will redirect to patient homepage	As expected	Passed
2	View License Dentists.	It’s responsive when you hover your mouse to the description box	As expected	Passed
3	View Personalized Treatment	It’s responsive when you hover your mouse to the description box	As expected	Passed
4	View Dental Goals	It’s responsive when you hover your mouse to	As expected	Passed

		the description box		
5	View Comfortable Center	It's responsive when you hover your mouse to the description box	As expected	Passed

Test Case ID:	TC_02

Test Scenario:	Comprehensive Dental Offerings "About"

Step #	Step Details	Expected Results	Actual Results	Remarks
1	Click the "About" navigation	You will redirect to about us page	As expected	Passed
2	View the quality oral wellness that dental clinics offer.	It's easy to read in bullet points and in different color font	As expected	Passed
3	Click the "Discover More" button	The other quality that dental clinics offer will show.	As expected	Passed

Test Case ID:	TC_03

Test Scenario:	Patients concern called "Contact us"

Step #	Step Details	Expected Results	Actual Results	Remarks
1	Click the “Contact Us” navigation	You will redirect to contact us page	As expected	Passed
2	View the contact information of the dental clinic.	It’s easy to understand the contact information because of the icon	As expected	Passed
3	Click the Map	You can see the exact location of the dental clinic.	As expected	Passed

Test Case ID:	TC_04

Test Scenario:	“Appointment” form for patients

Step #	Step Details	Expected Results	Actual Results	Remarks
1	Click the “Book Appointment” navigation	You will redirect to Appointment Form	As expected	Passed
2	Input all textboxes.	You need to input all the textboxes to save the data.	As expected	Passed
3	Click the SUBMIT button.	Your data will be saved, and you will have an appointment.	As expected	Passed

DOCTOR/STAFF PAGE

S#	Perquisite
1	Desktop/Laptop
2	Internet/Wifi
3	Browser/link

Test Case ID:	TC_01

Test Scenario:	Create Account

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In login form click the SIGN UP a text button.	The SIGNUP page will open.	As expected	Passed
2	Input all textboxes.	You need to input all the textboxes to save the data.	As expected	Passed
3	Click the SUBMIT button.	Your data will be saved, and you will have an account.	As expected	Passed
4	Click the SIGN IN a text button.	The SIGN IN page will open.	As expected	Passed

Test Case ID:	TC_02

Test Scenario:	Login form

Step #	Step Details	Expected Results	Actual Results	Remarks
1	Go to the website ”http://dentalapp.rf.gd/system/”	The Login page will open on the website.	As expected	Passed
2	Input the two textboxes.	Can input an Email Address and Password	As expected	Passed
3	Click the SUBMIT button.	The Dashboard page will open.	As expected	Passed
4	Click the SIGN UP text button.	The SIGNUP page will open.	As expected	Passed

Test Case ID:	TC_03

Test Scenario:	Dashboard can view total count record with time

Step #	Step Details	Expected Results	Actual Results	Remarks
1	View all Doctor’s menu	Every item that can be selected is functioning.	As expected	Passed
2	Click every item on Doctor’s menu	Every item that you click will have its own function.	As expected	Passed
3	View all Count Record	Can view how many total records	As expected	Passed
4	View Pie Graph’s total Men and Female	Can view both number and percentage of men and female	As expected	Passed
5	View Time	Can view exact time	As expected	Passed

Test Case ID:	TC_04

Test Scenario:	List of Doctor/Staff Account

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item a Doctor.	The Doctor page will open	As expected	Passed
2	View Doctor Account	Can view all Doctor Account without Email and Password in table	As expected	Passed
3	View Doctor Status	Can view Doctor Status in table	As expected	Passed
4	Search bar	The search bar is not working.	As expected	Failed
5	Show total row	Can view a total row	As expected	Passed
6	Click Previous and Next button	If you click the next button, the contents of the table are shown differently.	As expected	Passed

Test Case ID:	TC_05

Test Scenario:	Update Profile

Step #	Step Details	Expected Results	Actual Results	Remarks
1	On the header, in the name and	The Profile Page will open.	As expected	Passed

	picture, click the down arrow button, then click the profile.			
2	Click Image	Can click an image, then drop a file, or browse and save.	As expected	Passed
3	Input or update textbox	Can input or update a doctor's account.	As expected	Passed
4	Input a new or updated password.	Can input a new or updated doctor password.	As expected	Passed
5	Click Save Update button	If you can click the save update button, then your account is updated.	As expected	Passed

Test Case ID:	TC_06

Test Scenario:	Appointments form for doctor

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item Appointments.	The Appointments Page will open.	As expected	Passed
2	Input all textboxes.	You need to input all the textboxes to save the data.	As expected	Passed
3	Click the Save button.	Your data will be saved, and you	As expected	Passed

		will have an appointment.		
4	Click the Clear button.	All your input in the textbox will be erased.	As expected	Passed

Test Case ID:	TC_07

Test Scenario:	Slot Availability for schedule of patent

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item Slot Availability.	The Appointments Slot Page will open	As expected	Passed
2	View Schedule Patient	Can view of schedule patient in Callander	As expected	Passed
3	Edit input and reschedule the date and time for the patient.	Can edit and reschedule patients' date and times when the doctor is not available during the day.	As expected	Passed
4	Click the Resched button.	If you can click the Resched button, then your data is updated.	As expected	Passed

Test Case ID:	TC_08

Test Scenario:	Prescription of patent

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item Prescription.	The Prescription Slot Page will open	As expected	Passed
2	View record of patient	Can view record of patient in table	As expected	Passed
3	In tools on table click the button	You can click the button in Tools on the table, then show the information about the patient, name, doctor, date, time, and doctor ID of the prescription.	As expected	Passed
4	Search Bar	Can search the bar in the patient's name.	As expected	Passed
5	Click Previous and Next button	If you click the next button, the contents of the table are shown differently.	As expected	Passed
6	Click the Print button.	If you can click the Print button, then the prescription for the patient will be printed or saved in a pdf.	As expected	Passed

Test Case ID:	TC_09

Test Scenario:	Payment of patent

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item Payment .	The Payment Slot Page will open	As expected	Passed
2	View record of patient	Can view record of patient in table	As expected	Passed
3	In tools on table click the two button	You can click the payment processes button and print button the Tools on the table.	As expected	Passed
4	Click the payment processes button.	If you click the payment processes button, then you can input a description, price, quantity, and cash for the patient and then save.	As expected	Passed
5	Click the Print button.	If you can click the Print button, then the info and payment processes for the patient will be printed or saved in a pdf.	As expected	Passed
6	Click Previous and Next button	If you click the next button, the contents of the table are shown differently.	As expected	Passed

Test Case ID:	TC_10

Test Scenario:	Record of patent

Step #	Step Details	Expected Results	Actual Results	Remarks
1	In the Doctor's Menu, click item Payment .	The Payment Slot Page will open	As expected	Passed
2	View record of patient	Can view records of patients in the table when the patients have paid.	As expected	Passed
3	Click Previous and CSV and Excel text buttons	If you can click the CSV and Excel text buttons, then save in Microsoft Excel.	As expected	Passed
4	Click Previous and PDF text button	If you can click the PDF text button, then save in PDF then show the browser.	As expected	Passed
5	Click Previous and Print text button	If you can click the print text button, then show the browser, then print.	As expected	Passed
6	Search Bar	Can search bar in the patient's name	As expected	Passed
7	Click Previous and Next button	If you click the next button, the	As expected	Passed

		contents of the table are shown differently.		

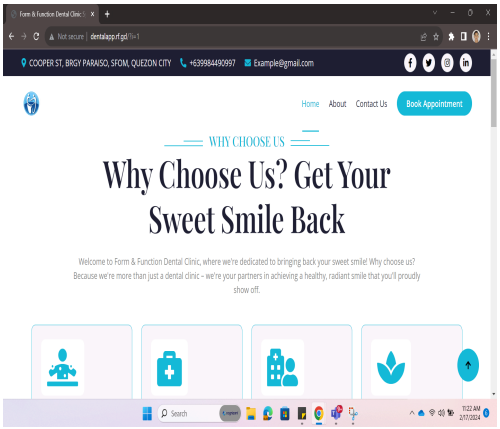
Test Case ID:	TC_11

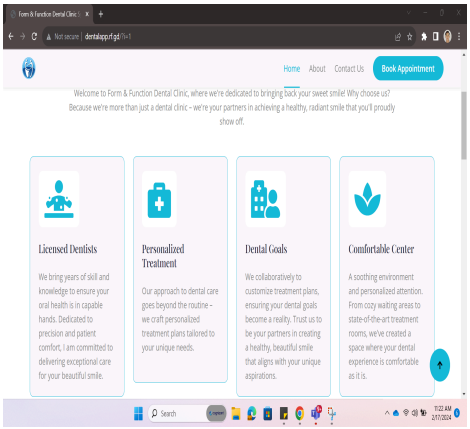
Test Scenario:	Log out

Step #	Step Details	Expected Results	Actual Results	Remarks
1	On the header, in the name and picture, click the down arrow button, then click the profile.	The SIGN IN page will open.	As expected	Passed

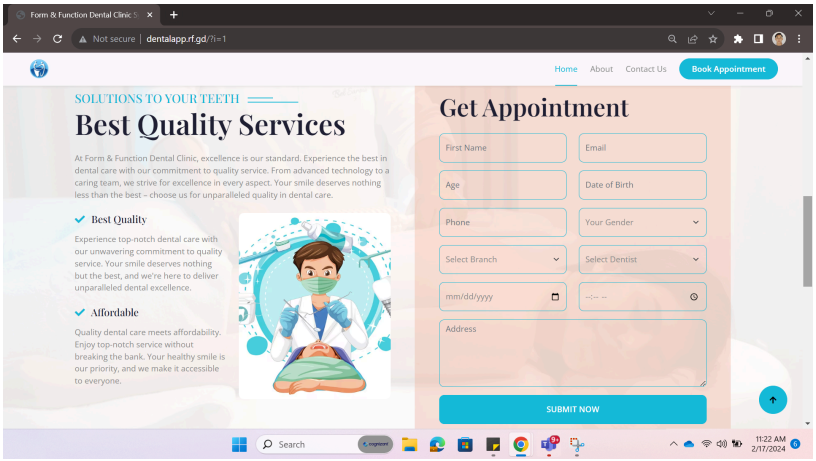
USER MANUAL

This user manual will walk them through the app's features and functions, allowing them to explore and make an appointment to the Form & Function Dental Clinic System.

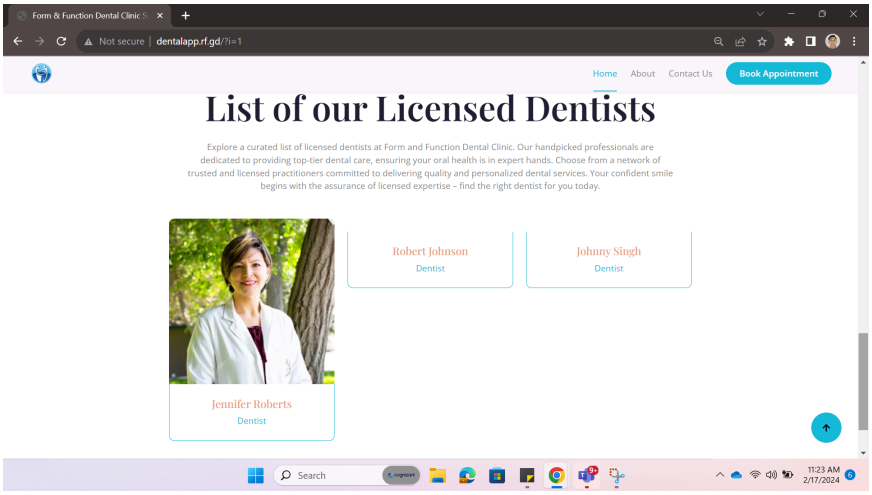




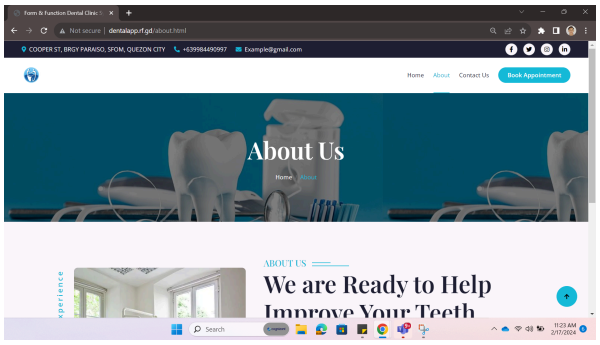
User Dashboard: The website will be sent to the dashboard when clicking on this link or visiting the website, where they can view all of the available information and further offers from the Dental Clinic

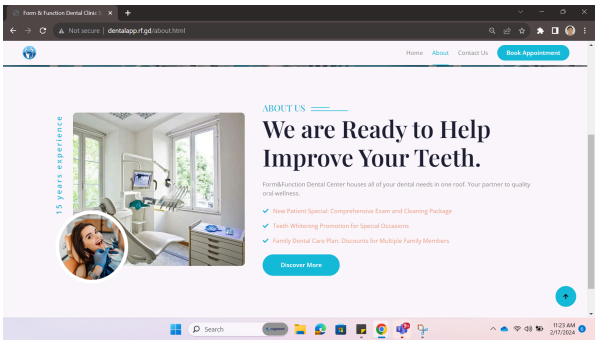


Get Appointment: They only need to scroll down to see where they may enter their personal information to make an appointment on the dashboard or home page.

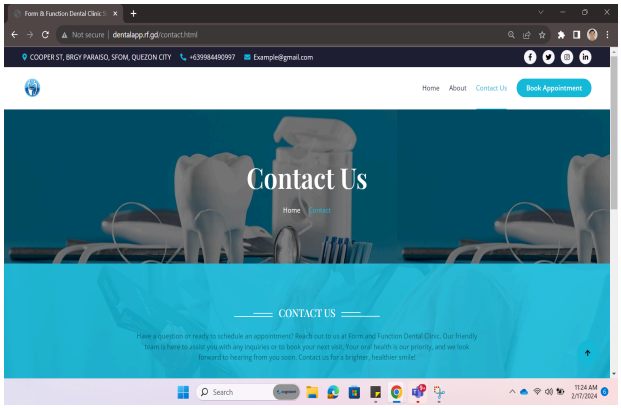


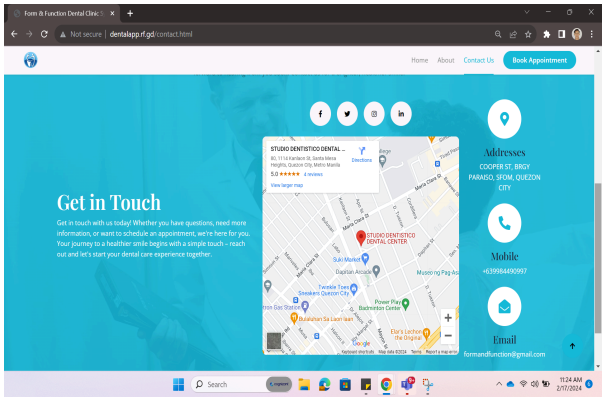
List of Licensed Dentists: Here are the listings of registered dentists with our clinic; they may also request a dentist by contacting the clinic.



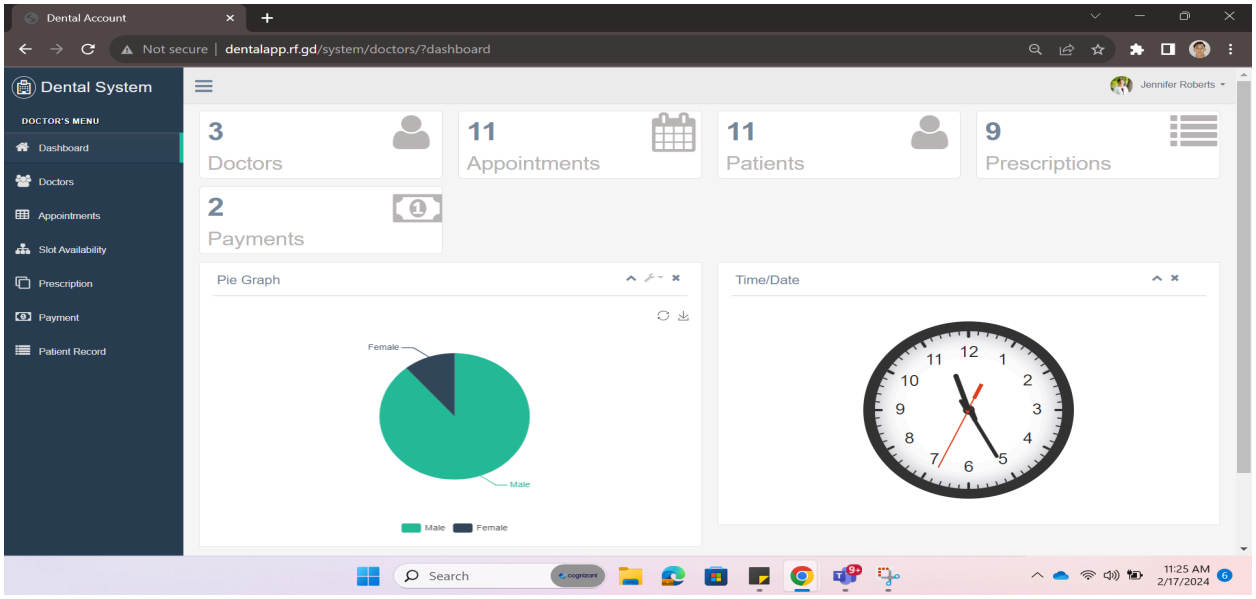


About Us: On the About Us page, they can see all the clinic's information and the offers, in case they are new and have yet to hear of the promotion our clinic has been running.

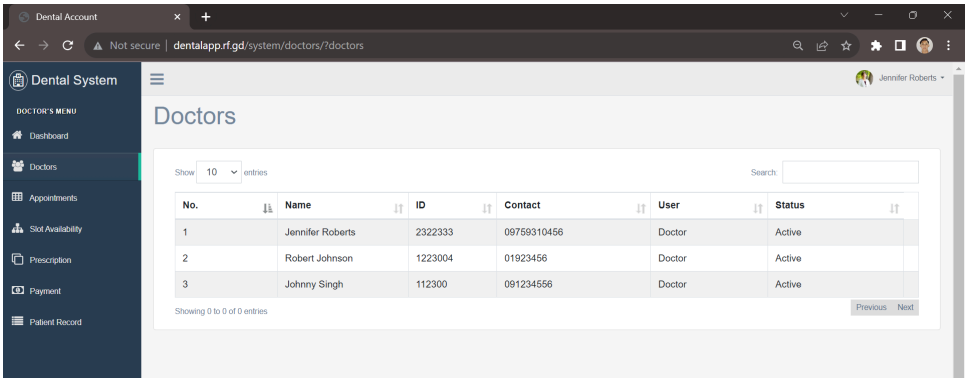




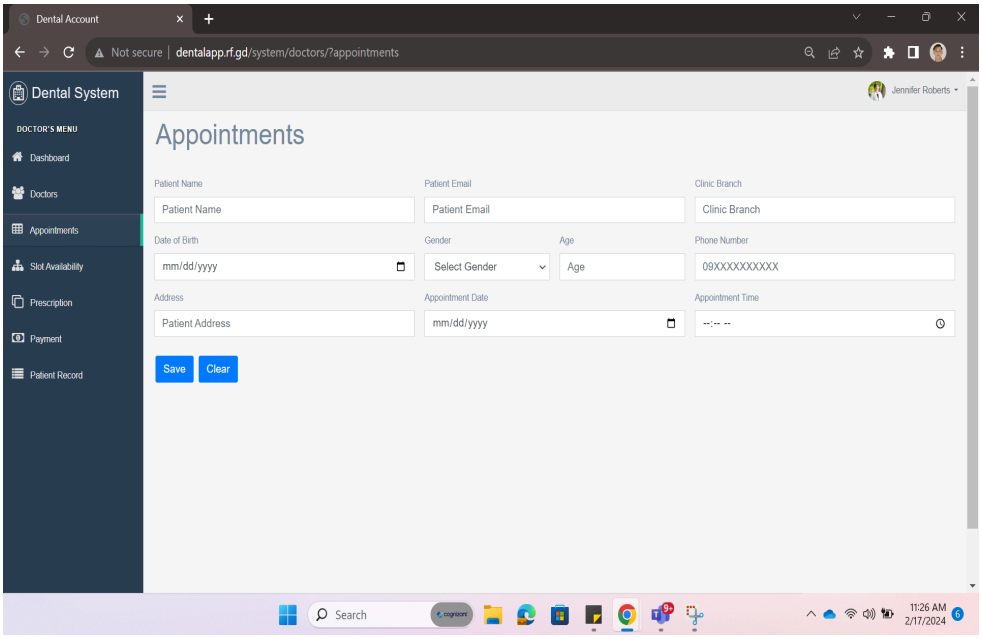
Contact Us: Click the "Contact Us" button on the home page if you have any questions or concerns about this page, including requests for feedback, promotions, or other matters.



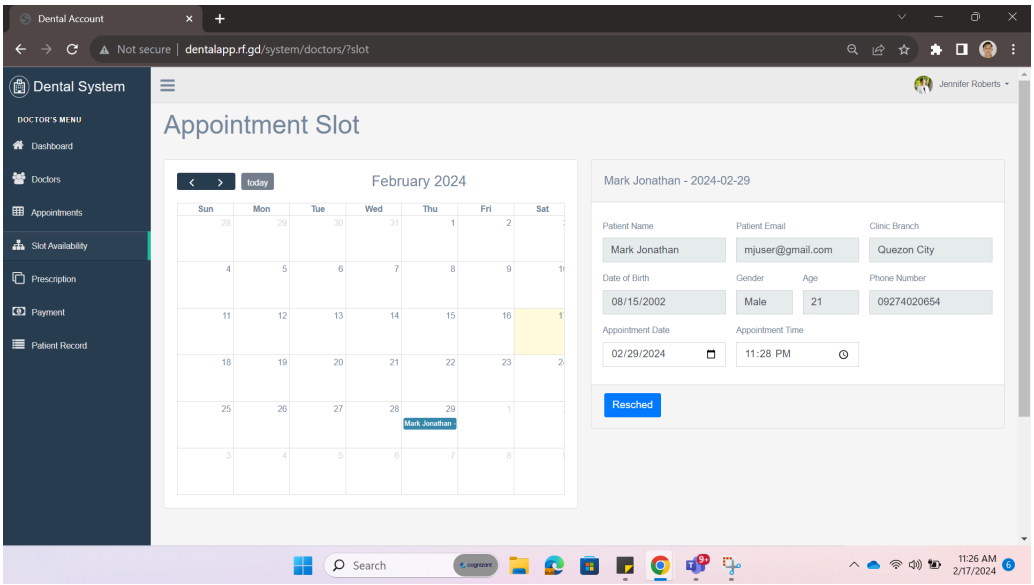
Doctor / Admin Dashboard: Once the doctor logs into their account, they will be taken to this page and can see all the information, such as if there are new appointments or patients, etc.



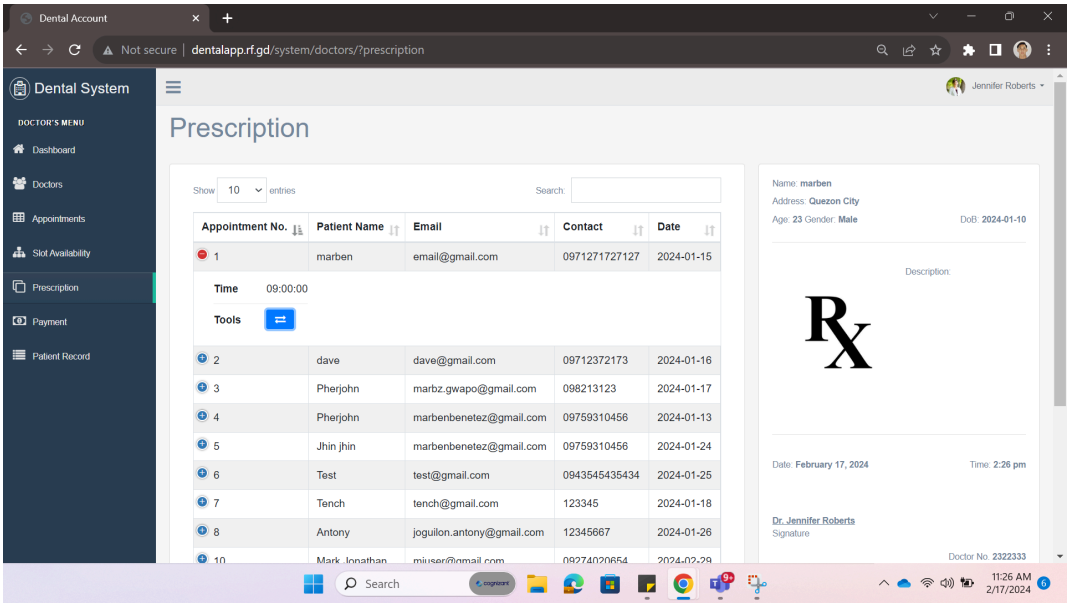
Doctors Page: These are the listings of doctors currently registered and affiliated with the clinic.



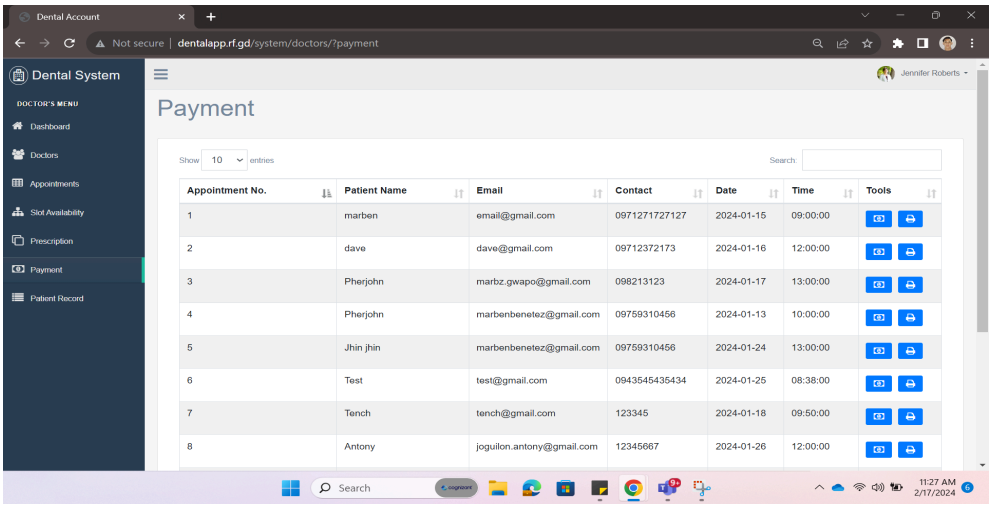
Appointments: We have made a method on this website for the doctor to manually schedule an appointment for clients who are unable to do so online or who are elderly and unfamiliar with the internet so they can schedule easily.



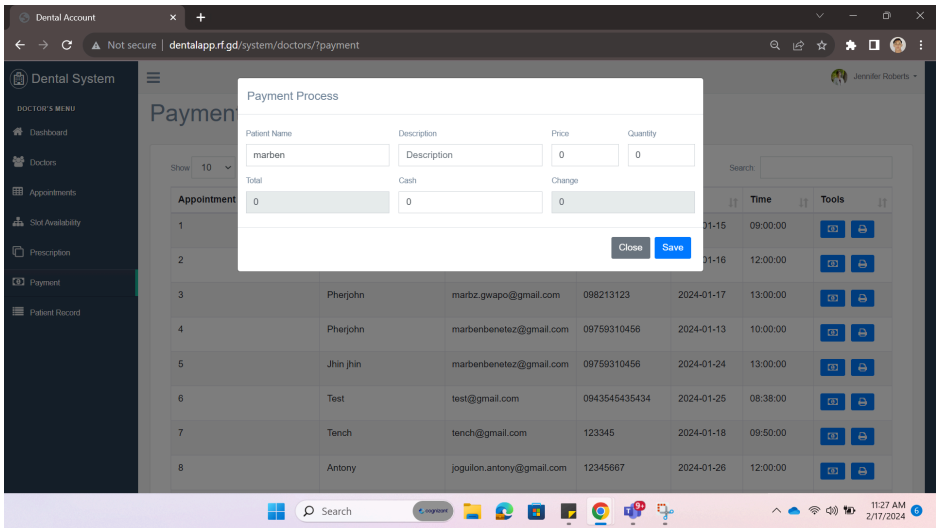
Appointment Slot: Dentists may check who has an appointment this month on this page. They can also reschedule a patient based on availability, but they must manually notify the patient.



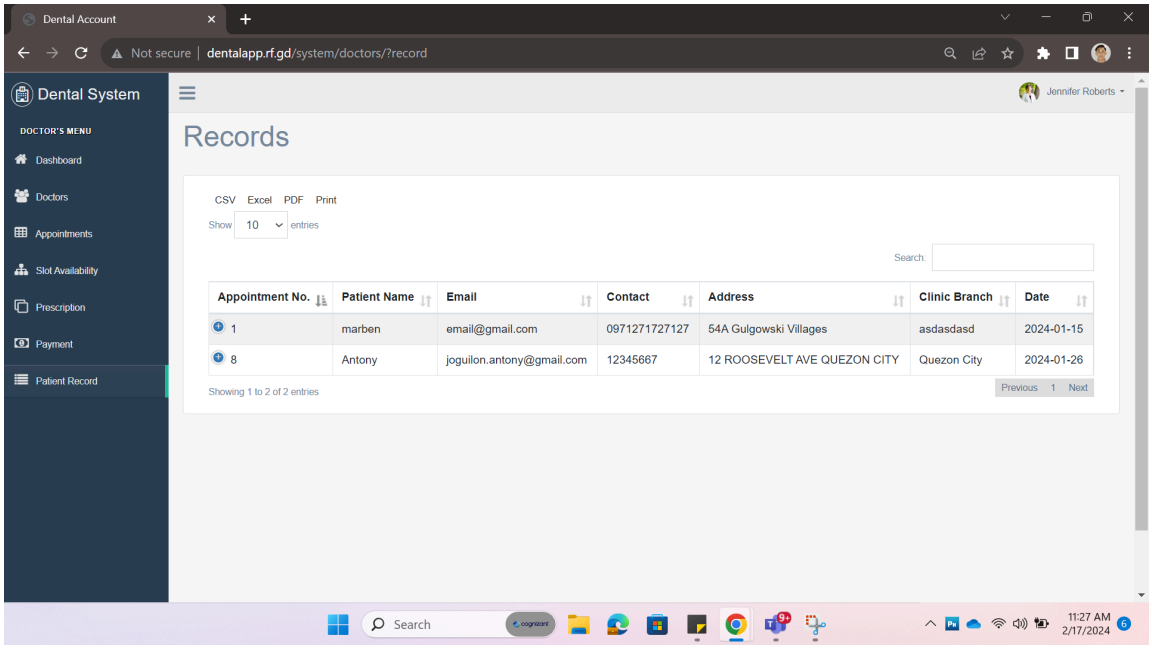
Prescription: If a prescription is needed after the appointment, the dentist will provide it to the patient so that it is still stored in the system, and they may print it instead of writing it down by hand



Payment: This is the payment portion, where the dentist will detail every procedure they have performed and the associated cost. The patient will then print the receipt.



Here is the sample on how to process the payment



Records: After completing the payment and process, they will be taken to this page on the clinic's records, where all previously completed information will be preserved and available for the dentists to review.

CERTIFICATIONS





CERTIFICATE

OF COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

FORM & FUNCTION DENTAL CLINIC SYSTEM

FOR COMPLETING THE
INFORMATION SECURITY ANALYST COURSE

ALLEN FAJARDO
COURSE FACILITATOR

ALEX CRUZ
ACADEMY PRESIDENT

INSTITUTE OF ARTS AND SCIENCES
Department of Communication

PLAGIARISM CHECK RESULT

On February 19, 2024, a plagiarism check was conducted on the system study entitled **"Form and Function Dental Clinic Integrated System."** The results showed an outstanding originality score of **96%,** confirming the authenticity and creativity of the work. The study has passed the plagiarism check, indicating that it is free from copied content.

Issued by:

Jeffrey Christian A. Lapeña

Grammarian

Far Eastern University – Manila

Date of Issuance:

February 21, 2024

INSTITUTE OF ARTS AND SCIENCES
Department of Communication

GRAMMARIAN CERTIFICATION

This is to certify that the undersigned has reviewed and went through all the pages of a system study entitled **"Form and Function Dental Clinic Integrated System"** by Garcia, John Royce S., Joguilon, Antony M., Quijano, Leonor F., Reytos, Mark Jonathan E., Siscon, Annie A., & Valenzuela, Franzh Yron., aligned with the set of structural rules that govern the composition of sentences, phrases, and words in the English language. Also, all corrections and recommendations have been made in/or incorporated in the final manuscript!

In view thereof, I have affixed my signature below as confirmation of the approval I have made hereto. Issued this on the 19th day of February, 2024.

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Far Eastern University - Manila

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